Data Communication & Computer Networks (CN)

Topic 1 of 7 : Introduction of DC (Data Communication)

113 MCQs — Repo Exam Questions + Forouzan — bilingual (English + Hindi), exam-ready

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **113** MCQs + **21** Nukte rapid-fire revision points

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter...)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

Syllabus Coverage — sub-topic wise order

1. Section A - DC Components & Protocol Basics (Message, Sender, Receiver, Transmission Medium, Protocol) — Q1 - Q16

2. Section B - Characteristics (Delivery, Accuracy, Timeliness, Jitter) & Criteria (Performance, Reliability, Security) — Q17 - Q30

3. Section C - Transmission Mode (Direction of Data Flow: Simplex, Half-Duplex, Full-Duplex) — Q31 - Q39

4. Section D - Network Structure / Topology (Physical: Mesh, Star, Bus, Ring, Others; Logical: IEEE 802) — Q40 - Q58

5. Section E - Types of Networks (LAN, MAN, WAN, PAN, CAN, SAN; Internet, Intranet, Extranet; Point-to-Point, Multipoint; Client-Server, P2P) — Q59 - Q90

6. Section F - Network History (ARPANET to Today's Network) — Q91 - Q98

7. Section G - General Network Basics (WWW, E-mail, Protocols, Servers, Switching - from repo intro tests) — Q99 - Q113

How to read: Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

System: generated by **tools/cn_build_topic.py** from **tools/data/topic1_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

Section A - DC Components & Protocol Basics (Message, Sender, Receiver, Transmission Medium, Protocol)

Syllabus: DC Components - Message, Sender, Receiver, Transmission Medium, Protocol

Q1. Identify the five components of a data communication system.

Data communication system के पाँच components की पहचान कीजिए।

(A) Message, Sender, Receiver, Transmission medium, Protocol **(B)** Message, Router, Switch, Cable, Software
(C) Data, Sender, Receiver, Hub, TCP/IP **(D)** Message, Sender, Receiver, Software, Hardware

[Source: Forouzan Ch1 (Q1-1) — Practice Set]

✔ **Correct: (A) Message, Sender, Receiver, Transmission medium, Protocol** — Forouzan lists exactly five components — Message (the information), Sender, Receiver, Transmission medium (the physical path) and Protocol (the rules).

Forouzan में पाँच components हैं — Message, Sender, Receiver, Transmission medium और Protocol।

★ **Nukta:** 5 components: Message → Sender → Medium → Receiver + Protocol (rules). सबसे favourite exam question.

5 components: Message → Sender → Medium → Receiver + Protocol (rules)। Exam का favourite question।

Q2. The _____ is the physical path over which a message travels from sender to receiver.

_____ वह physical path है जिस पर message sender से receiver तक जाता है।

(A) Path / पथ **(B)** Medium
(C) Protocol **(D)** Route / रूट (path)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q35]

✔ **Correct: (B) Medium** — In data communication, the transmission medium is the physical path (twisted-pair, coaxial, fiber or radio waves) by which the message travels from sender to receiver. (Forouzan)

Data communication में transmission medium वह physical path है (twisted-pair, coaxial, fiber या radio waves) जिस पर message sender से receiver तक जाता है। (Forouzan)

★ **Nukta:** Medium = physical path between sender and receiver. Remember: Message -> Sender -> Medium -> Receiver.

Medium = sender और receiver के बीच का physical path. Yaad रखो: Message -> Sender -> Medium -> Receiver.

Q3. The information to be communicated in a data communication system is called the _____.

Data communication system में communicate की जाने वाली information _____ कहलाती है।

(A) Medium **(B)** Protocol
(C) Message / message (message) **(D)** Transmission / संचरण (transmission)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q38]

✔ **Correct: (C) Message** — Forouzan: The message is the data (information) to be communicated. It can be text, numbers, images, audio or video.

Forouzan: Message = communicate की जाने वाली data (information). यह text, numbers, images, audio या video हो सकती है।

★ **Nukta:** Message = the data/information to be communicated. 5 popular forms: text, numbers, images, audio, video.

Message = communicate की जाने वाली data. 5 popular forms: text, numbers, images, audio, video.

Q4. Which of the following shares the data of two computers?

निम्नलिखित में से कौन दो computers का data share (साझा) करता है?

(A) Library / पुस्तकालय **(B)** Network
(C) Grouping / समूह (grouping) **(D)** Integrated system / एकीकृत प्रणाली

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q13]

✔ **Correct: (B) Network** — A computer network is a set of devices (nodes) connected together mainly to share data, resources and information between computers.

Computer network = devices (nodes) का ऐसा connection जो mainly computers के बीच data, resources और information share करने के लिए होता है।

★ **Nukta:** Network = 2 or more devices connected to share data/resources.

Network = 2 या ज़्यादा devices का आपस में जुड़ना, data/resources share करने के लिए।

Q5. In a computer network, nodes are:
Computer network में nodes होते हैं:

(A) the computer that originates the data / जो computer data को originate (shurua) करता है **(B)** the computer that routes the data / जो computer data को route करता है **(C)** the computer that terminates the data / जो computer data को terminate (समाप्त) करता है **(D)** all of the mentioned / उपर्युक्त सभी (all of the above)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q30]

✔ **Correct: (D) all of the mentioned** — A node is any device connected to the network. It can originate (sender), route (router/switch) or terminate (receiver) data, so all of the mentioned is correct.

Node = network से जुड़ा कोई भी device. यह data को originate (sender), route (router/switch) या terminate (receiver) कर सकता है, इसलिए uparyukt सभी सही हैं।

★ **Nukta:** Node = any connected device of the network (computer, printer, router, etc.).

Node = network का कोई भी connected device (computer, printer, router, ityadi).

Q6. _____ refers to two characteristics: when data should be sent and how fast it can be sent.
_____ दो characteristics को बताता है: data कब भेजा jaye और कितनी तेज़ी से भेजा जा सकता है।

(A) Semantics / अर्थ (semantics) **(B)** Syntax / संरचना (syntax) **(C)** Timing / समय-निर्धारण (timing) **(D)** none of the above / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q44]

✔ **Correct: (C) Timing** — The three key elements of a protocol are syntax, semantics and timing. Timing refers to two characteristics: when data should be sent and how fast it can be sent. (Forouzan)

Protocol के 3 key elements हैं: syntax, semantics और timing. Timing do characteristics से जुड़ा है: data कब भेजा jaye और कितनी तेज़ी से भेजा जा सकता है। (Forouzan)

★ **Nukta:** Protocol के 3 elements = Syntax (format), Semantics (meaning), Timing (when & how fast).

Protocol के 3 elements = Syntax (format), Semantics (अर्थ), Timing (कब और कितनी तेज़ी से).

Q7. Consider the statements: S1: The physical medium of connection is known as a link. S2: The computers connected to it are known as nodes.
कथनों पर विचार करें: S1: connection का physical medium link कहलाता है। S2: इससे जुड़े computers nodes कहलाते हैं।

(A) Both S1 and S2 are false / दोनों S1 और S2 ग़लत हैं **(B)** S1 true but S2 false / S1 सही है लेकिन S2 ग़लत है **(C)** S1 false but S2 true / S1 ग़लत है लेकिन S2 सही है **(D)** Both S1 and S2 true / दोनों S1 और S2 सही हैं

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q2]

✔ **Correct: (D) Both S1 and S2 true** — In networking, the physical medium connecting devices is called a link, and the connected devices (computers) are called nodes. Both statements are true.

Networking में devices को जोड़ने वाला physical medium link कहलाता है, और जुड़े हुए devices (computers) nodes कहलाते हैं। दोनों कथन सही हैं।

★ **Nukta:** Link = physical medium; Node = connected device (computer).

Link = physical medium; Node = जुड़ा हुआ device (computer).

Q8. What are the main elements of a protocol? 1. Syntax 2. Semantics 3. Timing
किसी protocol के मुख्य तत्व (elements) क्या हैं? 1. Syntax 2. Semantics 3. Timing

(A) 1, 2 only / सिर्फ 1 और 2 **(B)** 1, 3 only / सिर्फ 1 और 3 **(C)** 2, 3 only / सिर्फ 2 और 3 **(D)** 1, 2, 3 only / 1, 2 और 3 सभी

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q3]

✔ **Correct: (D) 1, 2, 3 only** — A protocol has three key elements: syntax (data format/structure), semantics (meaning of the data) and timing (when to send and how fast).

Protocol के 3 key elements हैं: syntax (data का format/structure), semantics (data का अर्थ) और timing (कब भेजना है और कितनी तेज़ी से).

★ **Nukta:** Yaad rakho: Protocol = Syntax + Semantics + Timing.

Yaad rakho: Protocol = Syntax + Semantics + Timing.

Q9. Why are protocols needed in data communication?

Data communication में protocols की आवश्यकता क्यों होती है?

(A) To enable communication between different types of systems by providing common rules / अलग-अलग प्रकार की systems के बीच communication के लिए common rules देना **(B)** To increase the processing speed of the CPU / CPU की processing speed बढ़ाने के लिए **(C)** To convert every message into a common language / हर message को एक common language में बदलने के लिए **(D)** Protocols are optional — devices can communicate without rules / Protocols optional हैं — बिना rules के भी devices communicate कर सकती हैं

[Source: Forouzan Ch1 (Q1-11) — Practice Set]

✔ **Correct: (A) To enable communication between different types of systems by providing common rules** — An agreement (protocol) of rules is needed before communication can begin — without common rules, unlike devices cannot understand each other.

Communication से पहले rules का agreement (protocol) ज़रूरी है — बिना common rules के devices एक-दूसरे को समझ नहीं सकतीं।

★ **Nukta:** Protocol = agreement of rules (syntax + semantics + timing). Iske बिना communication impossible.

Protocol = rules का agreement (syntax + semantics + timing)। इसके बिना communication impossible है।

Q10. A set of rules that governs data communication is called a:

Data communication को regulate (नियंत्रित) करने वाले rules का set _____ कहलाता है:

(A) rule / नियम (rule) **(B)** medium / medium (medium) **(C)** Link / link (link) **(D)** Protocol / protocol (protocol)

[Source: Repo: CN Topic Test-03 (UNIT TEST-3), Q15]

✔ **Correct: (D) Protocol** — A protocol is a set of rules that governs data communication. It defines what is communicated, how it is communicated and when it is communicated. (Forouzan)

Protocol = नियमों का set जो data communication को govern करता है. यह बताता है की क्या communicate होगा, कैसे होगा और कब होगा। (Forouzan)

★ **Nukta:** Protocol = rules का set. Yaad रखो: without a protocol, two devices may be connected but cannot communicate.

Protocol = नियमों का set. Yaad रखो: बिना protocol के do devices जुड़े to हो सकते हैं लेकिन communicate नहीं कर सकते।

Q11. The physical path between the sender and receiver is called:

Sender और receiver के बीच का physical path _____ कहलाता है:

(A) Link / link (link) **(B)** Node / node (node) **(C)** Segment / सेगमेंट (segment) **(D)** Socket / सॉकेट (socket)

[Source: Repo: BCI_SCI Subject Test-7 (UNIT TEST-2 BCI/SCI), Q20]

✔ **Correct: (A) Link** — In this context, the physical medium of connection between devices is known as a link (the devices connected to it are called nodes).

Is context में, devices को जोड़ने वाला physical medium link कहलाता है (is से जुड़े devices को nodes कहते हैं).

★ **Nukta:** Note: In Forouzan's 5 components, this same physical path is called the transmission medium. Both terms mean the physical path.

Note: Forouzan के 5 components में इसी physical path को transmission medium कहा गया है। दोनों का मतलब physical path ही है।

Q12. Consider: (1) Twisted pair cables (2) Microwaves and satellite signals (3) Repeaters (4) Analog transmissions (5) Fiber optics. Which of the above is considered as signal transmission medium in data communications?

विचार करें: (1) Twisted pair cables (2) Microwaves और satellite signals (3) Repeaters (4) Analog transmissions (5) Fiber optics। उपरोक्त में से data communications में signal transmission medium के रूप में क्या माना जाता है?

(A) (1) and (5) / (1) और (5) **(B)** (1) and (2) / (1) और (2) **(C)** (1), (2) and (5) / (1), (2) और (5) **(D)** (1), (2), (3) and (5) / (1), (2), (3) और (5)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q10]

✔ **Correct: (C) (1), (2) and (5)** — Twisted pair, microwaves/satellite and fiber optics are transmission media (guided and unguided). A repeater is a device (it regenerates the signal), and analog transmission is a form of signal, not a medium.

Twisted pair, microwaves/satellite और fiber optics transmission media हैं (guided और unguided). Repeater एक device है (signal regenerate करता है), और analog transmission signal का एक रूप है, medium नहीं।

★ **Nukta:** Guided media: twisted pair, coaxial, fiber. Unguided media: radio, microwave, infrared.

Guided media: twisted pair, coaxial, fiber. Unguided media: radio, microwave, infrared.

Q13. Data communication is the exchange of _____ between two devices via some form of transmission medium.

Data communication = do devices के बीच _____ का exchange, किसी transmission medium के through।

(A) only text messages / सिर्फ text messages **(B)** data (the message) - text, numbers, images, audio or video / data (message) - text, numbers, images, audio या video
(C) only electrical signals / सिर्फ electrical signals **(D)** only hardware devices / सिर्फ hardware devices

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (B) data (the message) - text, numbers, images, audio or video** — Forouzan: Data communications is the exchange of data between two devices via some form of transmission medium. The data exchanged is called the message.

Forouzan: Data communications = do devices के बीच किसी transmission medium के through data का exchange. Exchange होने वाले data को message कहते हैं।

★ **Nukta:** Data communication = exchange of DATA (message) between two devices over a medium.

Data communication = do devices के बीच DATA (message) का exchange, medium के through।

Q14. Which one of the following is NOT one of the five basic components of a data communication system?

निम्नलिखित में से कौन data communication system के five basic components में से एक नहीं है?

(A) Message / Message (message) **(B)** Protocol / Protocol (protocol)
(C) Router / Router (router) **(D)** Transmission medium / Transmission medium (medium)

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (C) Router** — The five basic components are: message, sender, receiver, transmission medium and protocol. A router is a networking device (Network layer), not one of the five fundamental components.

Five basic components हैं: message, sender, receiver, transmission medium और protocol. Router एक networking device है (Network layer), five fundamental components में से नहीं।

★ **Nukta:** Yaad रखो 5 components: Message, Sender, Receiver, Transmission Medium, Protocol. Router = device, component नहीं।

Yaad रखो 5 components: Message, Sender, Receiver, Transmission Medium, Protocol. Router = device है, component नहीं।

Q15. The device that sends the data message in a data communication system is called the _____.

Data communication system में जो device data message भेजता है, use _____ कहते हैं।

(A) Receiver / Receiver (प्राप्तकर्ता) **(B)** Sender / Sender (प्रेषक)
(C) Protocol / Protocol (protocol) **(D)** Hub / Hub (hub)

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (B) Sender** — The sender is the device (computer, workstation, mobile phone, camera, TV, etc.) that sends the data message.

Sender वह device है (computer, workstation, mobile phone, camera, TV, ityadi) जो data message भेजता है।

★ **Nukta:** Sender = source device; Receiver = destination device.

Sender = source device; Receiver = destination device.

Q16. Which of the following best lists the popular forms of information (message) in data communication?

Data communication में information (message) के popular forms को सबसे achhe से कौन बताता है?

(A) Only text and numbers / सिर्फ text और numbers **(B)** Only audio and video / सिर्फ audio और video
(C) Text, numbers, images, audio and video / text, numbers, images, audio और video **(D)** Only binary digits / सिर्फ binary digits

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (C) Text, numbers, images, audio and video** — Forouzan: The popular forms of information are text, numbers, images, audio, and video.

Forouzan: Information के popular forms हैं text, numbers, images, audio, और video.

★ **Nukta:** 5 popular forms of message: text, numbers, images, audio, video.

Message के 5 popular forms: text, numbers, images, audio, video.

Section B - Characteristics (Delivery, Accuracy, Timeliness, Jitter) & Criteria (Performance, Reliability, Security)

Syllabus: Characteristics - Delivery, Accuracy, Timeliness, Jitter | Properties/Criteria - Performance, Reliability, Security

Q17. What does jitter refer to in data communication?

Data communication में jitter का मतलब क्या होता है?

(A) Loss of data / data का loss **(B)** Delay variation in packets / packets में delay का अंतर (variation)
(C) Data synchronization / data का synchronization
(D) Data encoding / data का encoding

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q1]

✔ **Correct: (B) Delay variation in packets** — Jitter is the variation in packet arrival time. It is the uneven delay in the delivery of audio or video packets (e.g., video packets sent every 30 ms but arriving with 30 ms and 40 ms delays). (Forouzan)

Jitter = packets के पहुँचने के समय में अंतर (variation). यह audio या video packets की delivery में uneven delay है (जैसे video packets हर 30 ms पर bheje जाते हैं लेकिन 30 ms और 40 ms के delay से pahunchte हैं). (Forouzan)

★ **Nukta:** Jitter = uneven delay in packet arrival -> distorted audio/video playback.

Jitter = packet पहुँचने में uneven delay -> audio/video playback ख़राब हो जाती है।

Q18. Which of the following is a characteristic of data communication?

निम्नलिखित में से कौन data communication की एक characteristic (विशेषता) है?

(A) Timeliness / समयपालन (timeliness) **(B)** Delay / देरी (delay)
(C) Speed / गति (speed) **(D)** Size / आकार (size)

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q6]

✔ **Correct: (A) Timeliness** — The four fundamental characteristics of effective data communication are delivery, accuracy, timeliness and jitter. Timeliness means data must be delivered in a timely manner; data delivered late is useless.

Effective data communication की 4 fundamental characteristics हैं: delivery, accuracy, timeliness और jitter. Timeliness का मतलब है data time पर pahunchna; देर से पहुँचने वाला data बेकार है।

★ **Nukta:** 4 characteristics याद रखो: Delivery, Accuracy, Timeliness, Jitter.

4 characteristics याद रखो: Delivery, Accuracy, Timeliness, Jitter.

Q19. Accuracy in data communication means:

Data communication में accuracy का अर्थ है:

(A) Data is sent quickly / data तेज़ी से भेजा जाता है **(B)** Data is not corrupted / data corrupted (खराब) नहीं होता
(C) Data is encrypted / data encrypt होता है **(D)** Data reaches the receiver / data receiver तक pahunchta है

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q8]

✔ **Correct: (B) Data is not corrupted** — Accuracy means the system must deliver the data accurately. Data that has been altered in transmission and left uncorrected is useless.

Accuracy का मतलब है system data को accurately (सही) pahunchaye. Jo data transmission में badal गया और sudhara नहीं गया, बेकार है।

★ **Nukta:** Accuracy = error-free data; altered/corrupted data is useless.

Accuracy = error-free data; badal गया/corrupt data बेकार है।

Q20. What is the maximum data rate of a transmission medium known as?

Transmission medium की अधिकतम data rate को क्या कहा जाता है?

(A) Bandwidth **(B)** Throughput
(C) Latency **(D)** Jitter

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q9]

✔ **Correct: (A) Bandwidth** — Bandwidth is the maximum data rate (transmission capacity) of a communication channel/medium, usually measured in bits per second (bps).

Bandwidth = किसी communication channel/medium की अधिकतम data rate (transmission capacity) है, जिसे आम तौर पर bits per second (bps) में मापा जाता है।

★ **Nukta:** Bandwidth = maximum capacity of the channel (in bps). Throughput = actual rate achieved (usually less).

Bandwidth = channel की अधिकतम capacity (bps में). Throughput = asli में हासिल होने वाली rate (आम तौर पर कम).

Q21. Frequency of failure and network recovery time after a failure are measures of the _____ of a network.

Failure की aavritti (frequency) और failure के बाद network recovery time kis cheez के measures हैं?

(A) Performance **(B)** Reliability
(C) Security **(D)** Feasibility

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q39]

✔ **Correct: (B) Reliability** — Reliability is measured by the frequency of failure (how often the network fails), the time it takes to recover after a failure, and robustness in a catastrophe. (Forouzan)

Reliability को failure की frequency (network कितनी baar fail होता है), failure के बाद recover होने का time, और catastrophe में robustness से मापा जाता है। (Forouzan)

★ **Nukta:** Reliability = failure frequency + recovery time + robustness. Performance = transit time, response time, throughput, delay.

Reliability = failure frequency + recovery time + robustness. Performance = transit time, response time, throughput, delay.

Q22. An unauthorized user is a network _____ issue.

An unauthorized user एक network _____ issue है.

(A) Performance **(B)** Reliability
(C) Security **(D)** All of the above / ऊपर Sabhi

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q40]

✔ **Correct: (C) Security** — Security means protecting data from unauthorized access, protecting data from damage, and ensuring confidentiality and data integrity. An unauthorized user is a security issue.

Security का मतलब है data को unauthorized access से bachana, damage से bachana, और confidentiality और data integrity sunishchit करना। Unauthorized user एक security issue है।

★ **Nukta:** Security = unauthorized access से bachav + confidentiality + integrity (CIA triad).

Security = unauthorized access से bachav + confidentiality + integrity (CIA triad).

Q23. What is the criteria to check the reliability of a network? 1. Downtime 2. Failure Frequency 3. Fault Tolerance 4. Security

Network की reliability check करने के criteria क्या हैं? 1. Downtime 2. Failure Frequency 3. Fault Tolerance 4. Security

(A) 1, 2, 3 only / सिर्फ 1, 2, 3 **(B)** 2, 3, 4 only / सिर्फ 2, 3, 4
(C) 1, 2, 4 only / सिर्फ 1, 2, 4 **(D)** 1, 2, 3, 4 only / 1, 2, 3, 4 सभी

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q1]

✔ **Correct: (A) 1, 2, 3 only** — Reliability is measured by downtime/frequency of failure, recovery time and fault tolerance (robustness). Security is a separate criterion of a network, not a reliability measure.

Reliability को downtime/failure frequency, recovery time और fault tolerance (robustness) से मापा जाता है। Security एक अलग criterion है, reliability measure नहीं।

★ **Nukta:** 3 network criteria: Performance, Reliability, Security. Reliability = failure frequency + recovery time + fault tolerance.

3 network criteria: Performance, Reliability, Security. Reliability = failure frequency + recovery time + fault tolerance.

Q24. What are the three criteria necessary for an effective and efficient network?

Effective और efficient network के लिए आवश्यक तीन criteria क्या हैं?

(A) Performance, Reliability, Security **(B)** Delivery, Accuracy, Timeliness
(C) Speed, Cost, Size **(D)** Message, Sender, Receiver

[Source: Forouzan Ch1 (Q1-2) — Practice Set]

✔ **Correct: (A) Performance, Reliability, Security** — Forouzan: an effective network must satisfy three criteria — Performance (throughput/delay), Reliability (failure frequency/downtime) and Security (protection from unauthorised access).

Forouzan: network के तीन criteria — Performance, Reliability और Security।

★ **Nukta:** Network के 3 CRITERIA = Performance, Reliability, Security. Data communication के 4 CHARACTERISTICS = Delivery, Accuracy, Timeliness, Jitter. दोनों को confuse मत करना!

Network के 3 criteria = Performance, Reliability, Security। Data communication के 4 characteristics = Delivery, Accuracy, Timeliness, Jitter।

Q25. Which physical layer impairment mainly causes jitter in VoIP traffic?

VoIP traffic में jitter का मुख्य kaaran कौन सा physical layer impairment है?

(A) Attenuation / attenuation (signal की shakti कम होना)
(B) Delay variation / delay का अंतर (variation)
(C) Distortion / distortion (vikriti) **(D)** Crosstalk

[Source: Repo: BCI Subject Test-8 (UNIT TEST-3 BCI/SCI), Q36]

✔ **Correct: (B) Delay variation** — Jitter is caused by delay variation: packets experience different queuing/transmission delays, so they arrive at uneven intervals, distorting real-time VoIP/video.

Jitter delay variation से होता है: packets को अलग-अलग queuing/transmission delays milte हैं, इसलिए वह uneven intervals पर pahunchte हैं, जिससे real-time VoIP/video ख़राब होता है।

★ **Nukta:** Jitter = delay variation. Attenuation = signal weakening; distortion = signal shape change; crosstalk = interference from nearby cables.

Jitter = delay variation. Attenuation = signal की shakti कम होना; distortion = signal का रूप badalna; crosstalk = पास की cables से interference.

Q26. What network property is achieved through the use of the Automatic Repeat reQuest (ARQ) mechanism?

Automatic Repeat reQuest (ARQ) mechanism से network की कौन सी property हासिल होती है?

(A) Reliability **(B)** Availability
(C) Confidentiality **(D)** Scalability

[Source: Repo: Random Test- Rajasthan PYQ (RPSC/RSSB)]

✔ **Correct: (A) Reliability** — ARQ (Automatic Repeat reQuest) is an error-control mechanism: lost/corrupted frames are retransmitted, so reliable data transfer is achieved. Reliability is thus the property achieved.

ARQ (Automatic Repeat reQuest) एक error-control mechanism है: khoye/corrupt frames दोबारा bheje जाते हैं, जिससे reliable data transfer होता है। इसलिए reliability property हासिल होती है।

★ **Nukta:** ARQ = error control -> Reliability. Flow control = efficient delivery (no overflow).

ARQ = error control -> Reliability. Flow control = efficient delivery (receiver का buffer overflow नहीं होना).

Q27. For effective data communication, the system must deliver data to the correct destination. This characteristic is called _____.

Effective data communication के लिए system को data सही destination पर पहुँचाना चाहिए। Is characteristic को _____ कहते हैं।

(A) Accuracy **(B)** Timeliness
(C) Delivery **(D)** Jitter

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (C) Delivery** — Delivery: the system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user. (Forouzan)

Delivery: system को data सही destination पर पहुँचाना चाहिए। Data सिर्फ usi intended device/user को pahunchna चाहिए जिस के लिए है। (Forouzan)

★ **Nukta:** 4 characteristics: Delivery (right destination), Accuracy (error-free), Timeliness (on time), Jitter (even delay).

4 characteristics: Delivery (सही jagah), Accuracy (error-free), Timeliness (time पर), Jitter (even delay).

Q28. Data delivered late is useless (especially in real-time audio/video). Delivering data in a timely manner is called _____.

देर से पहुँचने वाला data बेकार है (ख़ास कर real-time audio/video में)। Time पर data pahunchane को _____ कहते हैं।

(A) Delivery **(B)** Accuracy
(C) Timeliness **(D)** Jitter

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (C) Timeliness** — Timeliness: the system must deliver data in a timely manner. Data delivered late is useless. For video/audio, timely delivery means delivering data as they are produced, in the same order, without significant delay (real-time transmission). (Forouzan)

Timeliness: system को data time पर पहुँचाना चाहिए। देर से पहुँचने वाला data बेकार है। Video/audio के लिए timely delivery का मतलब है data जिस तरह produce हुए हैं usi तरह, बिना किसी बड़े delay के पहुँचाना (real-time transmission). (Forouzan)

★ **Nukta:** Timeliness = on-time delivery; late data is useless -> real-time transmission (audio/video).

Timeliness = time पर पहुँचाना; देर वाला data बेकार है -> real-time transmission (audio/video).

Q29. The network criterion measured by transit time, response time, throughput and delay is called _____.

Transit time, response time, throughput और delay से maape jane वाले network criterion को _____ कहते हैं।

(A) Performance **(B)** Reliability
(C) Security **(D)** Delivery

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (A) Performance** — Performance of a network is measured by transit time (time for a message to travel from one device to another), response time (time between request and response), throughput and delay. (Forouzan)

Network की performance को transit time (एक device से दूसरे तक message पहुँचने का time), response time (request और response के बीच का time), throughput और delay से मापा जाता है। (Forouzan)

★ **Nukta:** Performance = transit time + response time (throughput & delay are related measures).

Performance = transit time + response time (throughput और delay isse जुड़े measures हैं).

Q30. Which of the following is NOT one of the three fundamental criteria (properties) that a network must meet?

निम्नलिखित में से कौन network के तीन fundamental criteria (properties) में से एक नहीं है?

(A) Performance **(B)** Reliability
(C) Security **(D)** Topology

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (D) Topology** — Forouzan: a network must meet certain criteria; the most important are performance, reliability, and security. Topology is a structure/arrangement, not one of the three criteria.

Forouzan: एक network को कुछ criteria पूरे करने चाहिए; सबसे ज़रूरी हैं performance, reliability और security. Topology एक structure/arrangement है, तीन criteria में से नहीं।

★ **Nukta:** Network criteria = 3: Performance, Reliability, Security.

Network criteria = 3: Performance, Reliability, Security.

Section C - Transmission Mode (Direction of Data Flow: Simplex, Half-Duplex, Full-Duplex)

Syllabus: Transmission Mode - Simplex, Half-Duplex, Full-Duplex

Q31. Which of the following transmission directions listed is not a legitimate channel?

निम्नलिखित में से कौन सा transmission direction एक legitimate (vaidh) channel नहीं है?

(A) Simplex **(B)** Half Duplex
(C) Full Duplex **(D)** Double Duplex

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q3]

✔ **Correct: (D) Double Duplex** — The three legitimate directions of data flow are simplex (one way only), half-duplex (both ways, one at a time) and full-duplex (both ways simultaneously). Double duplex is not a stअंदरd term.

Data flow के तीन legitimate directions हैं: simplex (एक ही तरफ़), half-duplex (दोनों तरफ़, एक समय में एक), full-duplex (दोनों तरफ़, एक साथ). Double duplex कोई stअंदरd term नहीं है।

★ **Nukta:** 3 transmission modes: Simplex, Half-Duplex, Full-Duplex. Double duplex = गलत.

3 transmission modes: Simplex, Half-Duplex, Full-Duplex. Double duplex = गलत है।

Q32. Communication between a computer and a keyboard involves _____ transmission.

Computer और keyboard के बीच communication में _____ transmission होता है।

(A) Automatic **(B)** Half-duplex
(C) Half-duplex (variant) / half-duplex **(D)** Simplex

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q33]

✔ **Correct: (D) Simplex** — A keyboard only sends data to the computer; the computer does not send data back to the keyboard. So the communication is unidirectional = simplex.

Keyboard सिर्फ computer को data भेजता है; computer keyboard को data wapas नहीं भेजता. इसलिए communication एक ही direction में है = simplex.

★ **Nukta:** Keyboard -> Computer = simplex (one way). Examples: keyboard-monitor, TV broadcast.

Keyboard -> Computer = simplex (एक तरफ़). Examples: keyboard-monitor, TV broadcast.

Q33. A television broadcast is an example of _____ transmission.

Television broadcast _____ transmission का एक example है।

(A) simplex **(B)** half-duplex
(C) full-duplex **(D)** automatic

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q41]

✔ **Correct: (A) simplex** — TV broadcast is unidirectional: the TV station only transmits and the TV set only receives. One direction only = simplex.

TV broadcast एक direction का है: TV station सिर्फ transmit करता है और TV set सिर्फ receive करता है। Ek ही direction = simplex.

★ **Nukta:** TV/radio broadcast = simplex (station -> TV only, wapas नहीं).

TV/radio broadcast = simplex (station -> TV ही, wapas नहीं).

Q34. What is the mode of transmission where data flows in only one direction?

Wo transmission mode kaunsa है जिसमें data सिर्फ एक direction में flow करता है?

(A) Simplex **(B)** Half-Duplex
(C) Full-Duplex **(D)** Multiplex

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q3]

✔ **Correct: (A) Simplex** — Simplex: communication is unidirectional, like a one-way street. Only one of the two devices on a link can transmit; the other can only receive. (Forouzan)

Simplex: communication एक ही direction का होता है, जैसे one-way street. Link के do devices में से सिर्फ एक transmit कर सकता है; दूसरा सिर्फ receive करता है। (Forouzan)

★ **Nukta:** Simplex = one-way only (keyboard to monitor, TV broadcast).

Simplex = सिर्फ एक तरफ़ (keyboard to monitor, TV broadcast).

Q35. Which transmission mode is used in walkie-talkie communication?

Walkie-talkie communication में कौन सा transmission mode use होता है?

(A) Simplex **(B)** Half-Duplex
(C) Full-Duplex **(D)** Multiplex

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q10]

✔ **Correct: (B) Half-Duplex** — A walkie-talkie is the classic example of half-duplex: both sides can talk, but only one at a time (push-to-talk). (Forouzan)

Walkie-talkie half-duplex का classic example है: दोनों तरफ़ baat कर सकते हैं, लेकिन एक समय में सिर्फ एक (push-to-talk). (Forouzan)

★ **Nukta:** Walkie-talkie = half-duplex (push-to-talk, एक समय में एक ही bolega).

Walkie-talkie = half-duplex (push-to-talk, एक समय में एक ही bolega).

Q36. What is the difference between half-duplex and full-duplex transmission modes?

Half-Duplex और Full-Duplex transmission modes में अंतर क्या है?

(A) Half-duplex allows only one-way flow; full-duplex also allows only one-way flow / Half-Duplex में सिर्फ one-way flow; Full-Duplex में भी सिर्फ one-way flow **(B)** Half-duplex flows both ways one direction at a time; full-duplex flows both ways at the same time / Half-Duplex में data दोनों तरफ़ पर एक समय में एक direction; Full-Duplex में एक साथ दोनों तरफ़

(C) Half-duplex is only for wireless; full-duplex only for wired networks / Half-Duplex सिर्फ wireless के लिए; Full-Duplex सिर्फ wired के लिए **(D)** There is no difference between the two / दोनों में कोई अंतर नहीं है

[Source: Forouzan Ch1 (Q1-6) — Practice Set]

✔ **Correct: (B) Half-duplex flows both ways one direction at a time; full-duplex flows both ways at the same time** — Half-duplex = both directions, one at a time (walkie-talkie). Full-duplex = both directions simultaneously (telephone).

Half-Duplex = दोनों direction, पर एक समय में एक (walkie-talkie)। Full-Duplex = एक साथ दोनों direction (telephone)।

★ **Nukta:** Half = एक time पर एक ही side (walkie-talkie). Full = दोनों side एक साथ (telephone).

Half = एक समय में एक side (walkie-talkie)। Full = दोनों side एक साथ (telephone)।

Q37. Data flow between two devices can occur in a _____ way.

Do devices के बीच data flow _____ tareeke से हो सकता है।

(A) simplex **(B)** half-duplex
(C) full-duplex **(D)** all of the above / ऊपर Sabhi

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q45]

✔ **Correct: (D) all of the above** — Data flow between two devices can be simplex (one way), half-duplex (both ways, one at a time) or full-duplex (both ways at the same time). Hence all of the above.

Do devices के बीच data flow simplex (एक तरफ़), half-duplex (दोनों तरफ़, एक समय में एक) या full-duplex (दोनों तरफ़, एक साथ) हो सकता है। इसलिए ऊपर सभी सही हैं।

★ **Nukta:** 3 ways of data flow: simplex, half-duplex, full-duplex.

Data flow के 3 tareeke: simplex, half-duplex, full-duplex.

Q38. In a telephone conversation, both persons can speak and hear each other at the same time. This is an example of _____.

Telephone conversation में दोनों persons एक साथ bol और sun सकते हैं। यह _____ का example है।

(A) Simplex **(B)** Half-duplex
(C) Full-duplex **(D)** Unicast

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (C) Full-duplex** — Full-duplex: both stations can transmit and receive simultaneously. The telephone network is Forouzan's example of full-duplex.

Full-duplex: दोनों stations एक साथ transmit और receive कर सकते हैं। Telephone network full-duplex का Forouzan का example है।

★ **Nukta:** Telephone = full-duplex (दोनों एक साथ bol और sun सकते हैं).

Telephone = full-duplex (दोनों एक साथ bol और sun सकते हैं).

Q39. Which of the following statements are TRUE? 1. Hubs operate in Half-Duplex mode. 2. Switches operate in Full-Duplex mode. 3. Routers broadcast by default.

निम्नलिखित में से कौन से statements TRUE हैं? 1. Hubs Half-Duplex mode में काम करते हैं। 2. Switches Full-Duplex mode में काम करते हैं। 3. Routers by default broadcast करते हैं।

(A) 1 and 2 / 1 और 2 **(B)** 2 and 3 / 2 और 3
(C) 1 and 3 / 1 और 3 **(D)** 1, 2 and 3 / 1, 2 और 3

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (A) 1 and 2** — Statement 1 TRUE: a hub shares one collision domain and works in half-duplex. Statement 2 TRUE: a switch gives each port its own collision domain and supports full-duplex. Statement 3 FALSE: routers do NOT forward broadcasts by default (each router port = separate broadcast domain).

Statement 1 TRUE: hub एक collision domain share करता है, Half-Duplex में काम करता है। Statement 2 TRUE: switch हर port को अलग collision domain देता है, Full-Duplex support करता है। Statement 3 FALSE: routers by default broadcasts forward नहीं करते।

★ **Nukta:** Hub = half-duplex + एक collision domain; Switch = full-duplex + per-port collision domain; Router = broadcasts rokta है.

Hub = Half-Duplex + एक collision domain; Switch = Full-Duplex + per-port collision domain; Router = broadcasts रोकता है।

Section D - Network Structure / Topology (Physical: Mesh, Star, Bus, Ring, Others; Logical: IEEE 802)

Syllabus: Physical Topology - Mesh, Star, Bus, Ring, Others (Hybrid, Tree) | Logical Topology - IEEE 802

Q40. What are the advantages of star topology?

Star topology के क्या advantages (लाभ) हैं?

(A) Less number of cables are required / कम number में cables चाहिए **(B)** More number of cables are required / ज़्यादा number में cables चाहिए
(C) It is a distributed topology / यह एक distributed topology है **(D)** Security is high / security ज़्यादा है

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q1]

✔ **Correct: (A) Less number of cables are required** — In a star topology, each device has a dedicated point-to-point link only to a central hub, so far fewer cables are required than in a mesh. It is also easy to install and reconfigure.

Star topology में हर device का सिर्फ central hub से dedicated point-to-point link होता है, इसलिए mesh की tulna में kaafi कम cables चाहिए. इसे install और reconfigure करना भी आसान है।

★ **Nukta:** Star = hub at centre, dedicated links, fewer cables (vs mesh), easy to install. But if hub fails, whole network goes down.

Star = centre में hub, dedicated links, कम cables (mesh के muqable), install करना आसान. लेकिन hub fail हो गया toh पूरा network band हो जाता है।

Q41. Which type of topology is best suited for large businesses which must carefully control and coordinate the operation of distributed branch outlets?

Large businesses के लिए कौन सी topology सबसे उपयुक्त है, jahan distributed branch outlets के operation को carefully control और coordinate करना हो?

(A) Ring **(B)** Local area
(C) Hierarchical / hierarchical (tree) **(D)** Star

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q2]

✔ **Correct: (C) Hierarchical** — A hierarchical (tree) topology is best for large organisations with distributed branch outlets because it is a hierarchy of stars and allows central control and coordination at each level.

Hierarchical (tree) topology large organisations के लिए सबसे उपयुक्त है jahan distributed branch outlets hon, क्योंकि यह stars की hierarchy है और हर level पर central control/coordination की अनुमति देती है।

★ **Nukta:** Tree = hierarchical star. Large orgs with branches -> hierarchical/tree topology.

Tree = hierarchical star. Badi sansthaon में branches के लिए -> hierarchical/tree topology.

Q42. Which one of the following is not a network topology?

निम्नलिखित में से कौन एक network topology नहीं है?

(A) Star **(B)** Ring
(C) Bus **(D)** Peer to Peer

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q9]

✔ **Correct: (D) Peer to Peer** — Star, ring and bus are physical topologies. Peer-to-peer is a network architecture (how devices share resources), not a topology.

Star, ring और bus physical topologies हैं। Peer-to-peer एक network architecture है (devices कैसे resources share karenge), topology नहीं।

★ **Nukta:** Topologies = mesh, star, bus, ring, hybrid, tree. P2P = architecture (not topology).

Topologies = mesh, star, bus, ring, hybrid, tree. P2P = architecture है (topology नहीं).

Q43. The repeater is used in _____ topology.

Repeater का use _____ topology में होता है।

(A) Mesh Topology **(B)** Ring Topology
(C) Bus Topology **(D)** Star Topology

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q50]

✔ **Correct: (C) Bus Topology** — In a bus topology, one long backbone cable carries the signal over the whole length; a repeater is used to regenerate/amplify the signal so it can travel the long distance without degradation.

Bus topology में एक लंबा backbone cable पूरी length पर signal le जाता है; repeater का use signal को regenerate/amplify करने के लिए होता है ताकि वह लंबी दूरी बिना ख़राब हुए जा sake।

★ **Nukta:** Bus = long backbone -> signal weak हो जाता है -> repeater use होता है. (Ring में भी हर device repeater की तरह काम करता है.)

Bus = लंबा backbone -> signal kamzor हो जाता है -> repeater use होता है। (Ring में भी हर device repeater की तरह काम करता है.)

Q44. What type of topology connects each node in a closed loop?

कौन सी topology हर node को एक closed loop में जोड़ती है?

(A) Star **(B)** Mesh
(C) Ring **(D)** Bus

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q4]

✔ **Correct: (C) Ring** — In a ring topology, each device is connected to its two neighbours (one on each side), forming a closed loop. Data travels in one direction and each device regenerates the signal (token passing).

Ring topology में हर device अपने दोनों neighbours से जुड़ा होता है (एक दोनों तरफ़), जिससे एक closed loop बनता है। Data एक ही direction में जाता है और हर device signal को regenerate करता है (token passing).

★ **Nukta:** Ring = closed loop, one direction, token passing, each device = repeater.

Ring = closed loop, एक direction, token passing, हर device = repeater.

Q45. In a _____ topology, every device has a dedicated point-to-point link to every other device. For n devices, the number of links is n(n-1)/2 and each device needs n-1 I/O ports.

_____ topology में हर device का हर दूसरे device से dedicated point-to-point link होता है। n devices के लिए links की संख्या n(n-1)/2 होती है और हर device को n-1 I/O ports चाहिए।

(A) Bus **(B)** Star
(C) Ring **(D)** Mesh

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (D) Mesh** — Mesh: every device has a dedicated point-to-point link to every other device. Number of links = n(n-1)/2; each device needs n-1 I/O ports. (Forouzan)

Mesh: हर device का हर दूसरे device से dedicated point-to-point link होता है। Links की संख्या = n(n-1)/2; हर device को n-1 I/O ports चाहिए। (Forouzan)

★ **Nukta:** Mesh formula (EXAM favourite): links = n(n-1)/2, ports per device = n-1. E.g., 5 devices -> 10 links.

Mesh formula (EXAM favourite): links = n(n-1)/2, हर device के ports = n-1. Jaise 5 devices -> 10 links.

Q46. For n devices in a network, how many cable links are required in a mesh topology?

Network में n devices के लिए mesh topology में कितने cable links चाहिए?

(A) n **(B)** n - 1
(C) n(n - 1)/2 **(D)** n(n - 1)

[Source: Forouzan Ch1 (Q1-8) — Practice Set]

✔ **Correct: (C) n(n - 1)/2** — In a mesh every device has a dedicated point-to-point link to every other device, so links = n(n-1)/2 (each link is shared by two devices).

Mesh में हर device का हर दूसरे device तक dedicated point-to-point link होता है, इसलिए links = n(n-1)/2।

★ **Nukta:** Mesh cables = n(n-1)/2, ports/device = n-1. Star = n cables. Ring = n. Bus = 1 backbone.

Mesh cables = n(n-1)/2, ports/device = n-1। Star = n। Ring = n। Bus = 1।

Q47. Six devices are arranged in a mesh topology. How many cables are needed, and how many ports does each device need?

छह devices को mesh topology में जोड़ा गया है। कितनी cables चाहिए और हर device को कितने ports चाहिए?

(A) 15 cables, 5 ports each / 15 cables, हर device पर 5 ports **(B)** 30 cables, 6 ports each / 30 cables, हर device पर 6 ports
(C) 15 cables, 6 ports each / 15 cables, हर device पर 6 ports **(D)** 6 cables, 5 ports each / 6 cables, हर device पर 5 ports

[Source: Forouzan Ch1 (P1-3) — Practice Set (Problem)]

✔ **Correct: (A) 15 cables, 5 ports each** — Links = n(n-1)/2 = 6×5/2 = 15 cables. Each device connects to the other 5 devices, so each needs n-1 = 5 ports.

Links = n(n-1)/2 = 6×5/2 = 15 cables। हर device बाकी 5 devices से जुड़ता है, इसलिए ports = 5।

★ **Nukta:** Mesh formula: cables = n(n-1)/2, ports/device = n-1. Numerical में हमेशा yahi do numbers maange जाते हैं.

Mesh: cables = n(n-1)/2, ports/device = n-1। Numerical में यही दो numbers पूछे जाते हैं।

Q48. In which topology does the failure of a single cable or device NOT affect the rest of the network?

किस topology में एक cable या device के fail होने पर बाकी network प्रभावित नहीं होता?

(A) Bus **(B)** Ring
(C) Star **(D)** Mesh

[Source: Forouzan Ch1 (P1-4) — Practice Set (Problem)]

✔ **Correct: (D) Mesh** — In a mesh, redundant links exist — if one link/device fails, traffic takes another path, so the rest of the network keeps working. In bus/ring a cable failure breaks the whole network; in star the hub is a single point of failure.

Mesh में redundant links होती हैं — एक link/device fail होने पर traffic दूसरे रास्ते से चलता है। Bus/ring में cable failure पूरा network तोड़ देता है; star में hub single point of failure है।

★ **Nukta:** Mesh = most fault-tolerant (redundant paths) — इसलिए 'highest robustness' mesh में होती है.

Mesh = सबसे ज़्यादा fault-tolerant (redundant paths) — इसीलिए mesh most robust है।

Q49. In a _____ topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub (or switch).

_____ topology में हर device का सिर्फ एक central controller (hub/switch) से dedicated point-to-point link होता है।

(A) Mesh **(B)** Ring
(C) Star **(D)** Bus

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (C) Star** — Star: each device has a dedicated point-to-point link only to a central controller, usually called a hub. The hub relays frames to the destination. (Forouzan)

Star: हर device का सिर्फ एक central controller (usually hub) से dedicated point-to-point link होता है। Hub frames को destination तक relay करता है। (Forouzan)

★ **Nukta:** Star = hub/switch at centre; easy to install & reconfigure; but if hub fails, the WHOLE network goes down.

Star = centre में hub/switch; install और reconfigure करना आसान; लेकिन hub fail हो गया toh POORA network band हो जाता है।

Q50. In a _____ topology, one long cable acts as a backbone to link all the devices; devices connect via drop lines and taps, and terminators are needed at both ends of the backbone.

_____ topology में एक लंबा cable backbone की तरह सभी devices को जोड़ता है; devices drop lines और taps से connect होते हैं, और backbone के दोनों ends पर terminators चाहिए।

(A) Bus **(B)** Star
(C) Ring **(D)** Mesh

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (A) Bus** — Bus: one long cable acts as a backbone to link all the devices. Nodes are connected to the bus cable by drop lines and taps. Terminators at both ends absorb the signal to prevent reflection. (Forouzan)

Bus: एक लंबा cable backbone की तरह सभी devices को जोड़ता है। Nodes को bus cable से drop lines और taps के through जोड़ा जाता है। दोनों ends पर terminators signal को absorb करते हैं ताकि reflection na हो। (Forouzan)

★ **Nukta:** Bus = one backbone + drop lines + taps + terminators at both ends. Fault isolation is difficult; a backbone break stops everything.

Bus = एक backbone + drop lines + taps + दोनों ends पर terminators. Fault isolation मुश्किल है; backbone tootne से सब कुछ band हो जाता है।

Q51. Which network topology requires a multipoint connection?

किस network topology में multipoint connection की आवश्यकता होती है?

(A) Ring **(B)** Bus
(C) Star **(D)** Mesh

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (B) Bus** — In a bus topology one long cable acts as a backbone and three or more devices share it — a multipoint connection. Mesh, star and ring use point-to-point links.

Bus topology में एक long cable backbone बनती है और तीन+ devices उसे share करती हैं — यही multipoint connection है। Mesh, star, ring point-to-point links use करते हैं।

★ **Nukta:** Bus = multipoint (shared backbone). Mesh/Star/Ring = point-to-point links. Yahi difference exam में poochha जाता है.

Bus = multipoint (shared backbone)। Mesh/Star/Ring = point-to-point links।

Q52. In a _____ topology, each device is connected to its two neighbours, data travels in one direction around the loop, and each device regenerates (repeats) the signal - this is called token passing.

_____ topology में हर device अपने do neighbours से जुड़ा होता है, data loop में एक ही direction में जाता है, और हर device signal को regenerate (repeat) करता है - इसे token passing कहते हैं।

(A) Bus **(B)** Ring
(C) Star **(D)** Mesh

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (B) Ring** — Ring: each device is connected to its two neighbours. Data travels in one direction. Each device in the ring incorporates a repeater; it regenerates the bits and passes them along. Access is typically by token passing. (Forouzan)

Ring: हर device अपने do neighbours से जुड़ा होता है। Data एक ही direction में जाता है। Ring का हर device एक repeater की तरह काम करता है; यह bits को regenerate karke आगे pass करता है। Access आम तौर पर token passing से होता है। (Forouzan)

★ **Nukta:** Ring = loop, one direction, token passing, each device = repeater.

Ring = loop, एक direction, token passing, हर device = repeater.

Q53. A network in which two or more different topologies are combined (e.g., a star connected to another star) is called a _____ topology, while a _____ topology is a hierarchy of stars (a variation of the star topology).

जिसमें do या ज़्यादा अलग topologies combine हो (जैसे एक star दूसरे star से जुड़ा हो), use _____ topology कहते हैं, और _____ topology stars की hierarchy है (star topology का variation).

(A) Hybrid ; Tree **(B)** Tree ; Hybrid
(C) Mesh ; Ring **(D)** Bus ; Star

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (A) Hybrid ; Tree** — Hybrid = a combination of two or more different topologies (e.g., a star of stars). Tree = a hierarchical arrangement of stars, a variation of the star topology.

Hybrid = do या ज़्यादा अलग topologies का combination है (जैसे star of stars). Tree = stars की hierarchical arrangement है, star topology का एक variation.

★ **Nukta:** Hybrid = mix of topologies; Tree = star of stars (hierarchical star).

Hybrid = topologies का mix; Tree = star of stars (hierarchical star).

Q54. The _____ topology of a network defines how the devices are physically laid out/connected, whereas the _____ topology defines how the data actually flow / how the devices access the network to transmit.

Network की _____ topology बताती है की devices physically कैसे जुड़े हैं, jabki _____ topology बताती है की data asal में कैसे flow करता है / devices transmit करने के लिए network को कैसे access करते हैं।

(A) Logical ; Physical **(B)** Physical ; Logical
(C) Star ; Bus **(D)** Mesh ; Ring

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (B) Physical ; Logical** — Physical topology = the actual layout of devices and cables. Logical topology = the way data flows / how devices access the medium. Two networks can have the same logical topology but different physical topologies.

Physical topology = devices और cables का asli layout. Logical topology = data कैसे flow करता है / devices medium को कैसे access करते हैं। Do networks की logical topology same हो सकती है लेकिन physical अलग हो सकती है।

★ **Nukta:** Physical = कैसे जुड़े (cables); Logical = data कैसे flow करता है (access). Same logical, different physical possible.

Physical = कैसे जुड़े (cables); Logical = data कैसे flow करता है (access). Same logical, different physical हो सकता है।

Q55. The IEEE 802 stअंदरds (such as 802.3 for Ethernet, 802.5 for Token Ring, 802.11 for Wi-Fi) primarily define the _____ of LANs/MANs.

IEEE 802 stअंदरds (जैसे 802.3 Ethernet, 802.5 Token Ring, 802.11 Wi-Fi) primarily LANs/MANs की _____ define करते हैं।

(A) Physical and Data Link layers (including the logical topology / MAC) / Physical और Data Link layers (logical topology / MAC sahit) **(B)** Only the Application layer / सिर्फ Application layer
(C) Only IP addressing / सिर्फ IP addressing **(D)** Only WAN routing / सिर्फ WAN routing

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (A) Physical and Data Link layers (including the logical topology / MAC)** — IEEE 802 is the family of stअंदरds for LANs and MANs; it covers the Physical layer and the Data Link layer (split into LLC - 802.2 - and MAC sublayers). 802.3 = Ethernet, 802.5 = Token Ring, 802.11 = Wireless LAN. The logical topology (bus/ring) is a Data-Link/MAC concern.

IEEE 802 LANs और MANs के लिए stअंदरds का family है; यह Physical layer और Data Link layer को cover करता है (LLC - 802.2 - और MAC sublayers में बनता है). 802.3 = Ethernet, 802.5 = Token Ring, 802.11 = Wireless LAN. Logical topology (bus/ring) एक Data-Link/MAC concern है।

★ **Nukta:** IEEE 802 = LAN/MAN stअंदरds (Physical + Data Link). 802.3 Ethernet, 802.5 Token Ring, 802.11 Wi-Fi.

IEEE 802 = LAN/MAN stअंदरds (Physical + Data Link). 802.3 Ethernet, 802.5 Token Ring, 802.11 Wi-Fi.

Q56. IEEE 802.2 stअंदरd is related to which sub-layer?

IEEE 802.2 stअंदरd किस sub-layer से संबंधित है?

(A) MAC Sublayer **(B)** LLC Sublayer
(C) Physical Layer **(D)** Network Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (B) LLC Sublayer** — IEEE 802.2 defines the Logical Link Control (LLC) sub-layer of the Data Link layer; the MAC sub-layer is defined by IEEE 802.3 (Ethernet).

IEEE 802.2 Data Link layer का LLC (Logical Link Control) sub-layer define करता है; MAC sub-layer IEEE 802.3 (Ethernet) से जुड़ा है।

★ **Nukta:** 802.2 = LLC (upper sub-layer of Data Link). 802.3 = Ethernet/MAC, 802.5 = Token Ring, 802.11 = Wi-Fi.

802.2 = LLC। 802.3 = Ethernet/MAC, 802.5 = Token Ring, 802.11 = Wi-Fi।

Q57. Which physical topology offers the highest robustness (fault tolerance) because of multiple redundant paths, but is also the most expensive and hardest to install because of the huge amount of cabling?

कौन सी physical topology सबसे ज़्यादा robust (fault tolerant) है क्योंकि ismein कई redundant paths होते हैं, लेकिन सबसे महंगी और install करने में सबसे मुश्किल भी है क्योंकि बहुत ज़्यादा cabling चाहिए?

(A) Bus **(B)** Ring
(C) Star **(D)** Mesh

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (D) Mesh** — Mesh has a dedicated link between every pair of devices, so the failure of one link does not affect the others - it is the most robust. But n(n-1)/2 links make it the most expensive and hardest to install/wire.

Mesh में हर pair of devices के बीच dedicated link होता है, इसलिए एक link fail होने से दूसरे affected नहीं होते - यह सबसे robust है। लेकिन n(n-1)/2 links की wajah से यह सबसे महंगी और install करने में सबसे मुश्किल है।

★ **Nukta:** Mesh = most robust + most costly (n(n-1)/2 cables). Star = cheap & easy but hub is a single point of failure. Bus = cheapest, hardest fault isolation.

Mesh = सबसे robust + सबसे महंगी (n(n-1)/2 cables). Star = sasti और आसान लेकिन hub single point of failure है. Bus = सबसे sasti, fault isolation सबसे मुश्किल.

Q58. Four network segments in Bus topology are: A1–A2, B1–B2–B3, C1–C2 and D1–D2–D3. If A1, B1, C1 and D1 are interconnected in a mesh, how many mesh connections are required?

Bus topology में चार network segments हैं: A1–A2, B1–B2–B3, C1–C2 और D1–D2–D3। यदि A1, B1, C1 और D1 को mesh में जोड़ा जाए, तो कितने mesh connections चाहिए?

(A) 12 **(B)** 10
(C) 6 **(D)** 45

[Source: RPSC Programmer 2024 Paper-1 (with official key)]

✔ **Correct: (C) 6** — Only the four interconnecting nodes (A1, B1, C1, D1) are meshed: links = n(n-1)/2 = 4×3/2 = 6. (Matches the official RPSC key.)

सिर्फ चार interconnecting nodes (A1, B1, C1, D1) mesh हो रहे हैं: links = n(n-1)/2 = 4×3/2 = 6। (Official RPSC key से match.)

★ **Nukta:** Mesh formula n(n-1)/2 — लेकिन n सिर्फ meshed nodes है (4), पूरे network के hosts (10) नहीं!

Mesh formula n(n-1)/2 — पर n सिर्फ meshed nodes है (4), पूरे network के hosts (10) नहीं!

Section E - Types of Networks (LAN, MAN, WAN, PAN, CAN, SAN; Internet, Intranet, Extranet; Point-to-Point, Multipoint; Client-Server, P2P)

Syllabus: Types of Networks - Geographical (LAN, MAN, WAN, PAN, CAN, SAN) | Accessibility (Internet, Intranet, Extranet) | Architecture (Point-to-Point, Multipoint)

Q59. The term WAN stands for?

WAN का पूरा नाम (full form) क्या है?

(A) Wide Area Net **(B)** Wide Access Network
(C) Wide Area Network **(D)** Wide Access Net

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q11 / BCI_SCI Subject Test-01, Q11]

✔ **Correct: (C) Wide Area Network** — WAN = Wide Area Network. It spans a large geographical area - a country, continent or the whole world. The Internet is the largest WAN.

WAN = Wide Area Network. यह एक बड़े geographical area को span करता है - एक देश, continent या पूरी दुनिया। Internet सबसे बड़ा WAN है।

★ **Nukta:** WAN = Wide Area Network. Internet सबसे बड़ा WAN है.

WAN = Wide Area Network. Internet सबसे बड़ा WAN है।

Q60. Which network is suitable for a building?

किसी एक building के लिए कौन सा network उपयुक्त (suitable) है?

(A) WAN **(B)** LAN
(C) MAN **(D)** PAN

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q27]

✔ **Correct: (B) LAN** — A LAN (Local Area Network) is usually privately owned and covers a small geographical area - a single office, building, or campus. So a building -> LAN.

LAN (Local Area Network) आम तौर पर privately owned होता है और एक chhote geographical area - एक office, building, या campus - को cover करता है। इसलिए building के लिए LAN उपयुक्त है।

★ **Nukta:** LAN = Local Area Network - छोटा area (office/building/campus), private, high speed.

LAN = Local Area Network - छोटा area (office/building/campus), private, high speed.

Q61. Bluetooth is an example of:
Bluetooth इसका example है:

(A) personal area network / personal area network (PAN) **(B)** local area network / local area network (LAN)
(C) virtual private network / virtual private network (VPN) **(D)** none of the mentioned / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q31]

✔ **Correct: (A) personal area network** — Bluetooth connects devices around one person (mobile phone with earphones/headset). A network organized around an individual person is a PAN (Personal Area Network).
Bluetooth एक व्यक्ति के aas-पास के devices को जोड़ता है (mobile phone को earphones/headset से). Ek व्यक्ति के around organized network को PAN (Personal Area Network) कहते हैं।
★ **Nukta:** PAN = Personal Area Network - एक person के around (Bluetooth: phone <-> earphones).
PAN = Personal Area Network - एक person के around (Bluetooth: phone <-> earphones).

Q62. Which one of the following extends a private network across public networks?
निम्नलिखित में से कौन एक private network को public networks के paar (across) extend करता है?

(A) local area network / local area network (LAN) **(B)** virtual private network / virtual private network (VPN)
(C) enterprise private network **(D)** storage area network / storage area network (SAN)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q32]

✔ **Correct: (B) virtual private network** — A Virtual Private Network (VPN) extends a private network across a public network (e.g., the Internet) using tunneling and encryption, so data travels securely as if on a private network.
Virtual Private Network (VPN) एक private network को एक public network (जैसे Internet) के paar extend करता है, tunneling और encryption के through, ताकि data private network की तरह securely travel kare।
★ **Nukta:** VPN = Virtual Private Network - private network को public (Internet) पर extend करता है, securely.
VPN = Virtual Private Network - private network को public (Internet) पर extend करता है, securely.

Q63. _____ is a network that covers geographic areas that are larger, such as districts or cities.
_____ एक ऐसा network है जो बड़े geographical areas, जैसे districts या cities, को cover करता है।

(A) LAN **(B)** MAN
(C) WAN **(D)** None of these / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q49]

✔ **Correct: (B) MAN** — MAN (Metropolitan Area Network) covers a metropolitan area, usually a city - larger than a LAN but smaller than a WAN. Example: cable TV network of a city.
MAN (Metropolitan Area Network) एक metropolitan area, आम तौर पर एक city, को cover करता है - LAN से बड़ा लेकिन WAN से छोटा। Example: एक city का cable TV network.
★ **Nukta:** MAN = Metropolitan Area Network - city level (cable TV network).
MAN = Metropolitan Area Network - city level (cable TV network).

Q64. Which factor primarily decides whether a communication system is a LAN or a WAN?
कौन-सा factor मुख्य रूप से तय करता है कि communication system LAN है या WAN?

(A) The geographical area it spans / वह कितने geographical area में फैला है **(B)** The number of protocols it uses / वह कितने protocols use करता है
(C) The colour of the cables / Cables का रंग **(D)** The type of keyboard attached / जुड़े keyboard का प्रकार

[Source: Forouzan Ch1 (Q1-9) — Practice Set]

✔ **Correct: (A) The geographical area it spans** — Forouzan: a LAN covers a building/plant/campus (small area); a WAN spans states, countries or the whole world — geographical span is the primary factor.
Forouzan: LAN building/campus जैसे छोटे area को cover करता है; WAN states/countries/दुनिया तक फैला होता है — geographical span ही मुख्य factor है।
★ **Nukta:** Chhoti area = LAN, city = MAN, state/country/world = WAN. Faisla सिर्फ area का.
छोटा area = LAN, city = MAN, state/country/world = WAN। फैसला सिर्फ area का।

Q65. In terms of the size of the network, the correct order (ascending) is:

Network के size के hisaab से सही kram (badhta हुआ/ ascending order) क्या है?

(A) PAN, MAN, LAN, WAN **(B)** LAN, MAN, WAN, PAN
(C) PAN, LAN, MAN, WAN **(D)** LAN, PAN, MAN, WAN

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q42]

✔ **Correct: (C) PAN, LAN, MAN, WAN** — Ascending order of size: PAN (around a person, smallest) < LAN (building/campus) < MAN (city) < WAN (country/world, largest).

Size के hisaab से badhta kram: PAN (एक person के around, सबसे छोटा) < LAN (building/campus) < MAN (city) < WAN (देश/ दुनिया, सबसे बड़ा).

★ **Nukta:** Yaad रखो: PAN < LAN < MAN < WAN (छोटा से बड़ा).

Yaad रखो: PAN < LAN < MAN < WAN (छोटा से बड़ा).

Q66. Which type of network covers an entire city or campus?

कौन सा network पूरे एक city या campus को cover करता है?

(A) LAN **(B)** MAN
(C) WAN **(D)** PAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q2]

✔ **Correct: (B) MAN** — MAN (Metropolitan Area Network) covers an entire city; a large campus can also be of MAN scale. (Official answer key: MAN)

MAN (Metropolitan Area Network) एक पूरे city को cover करता है; एक बड़ा campus भी MAN scale का हो सकता है। (Official answer key: MAN)

★ **Nukta:** MAN = city level. Note: a single building/ campus LAN is usually LAN/CAN; an entire city = MAN.

MAN = city level. Note: एक building/campus का LAN आम तौर पर LAN/CAN होता है; पूरी city = MAN.

Q67. Which of these is NOT a type of network?

इनमें से कौन network का एक prakar (type) नहीं है?

(A) LAN **(B)** MAN
(C) WAN **(D)** TAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q7]

✔ **Correct: (D) TAN** — LAN, MAN and WAN are stअंदरd network types (by geographical area). TAN is not a stअंदरd type of network.

LAN, MAN और WAN stअंदरd network types हैं (geographical area के aadhar पर). TAN कोई stअंदरd network type नहीं है।

★ **Nukta:** Network types: PAN, LAN, MAN, WAN (+ CAN, SAN). TAN = नहीं है.

Network types: PAN, LAN, MAN, WAN (+ CAN, SAN). TAN = नहीं है।

Q68. The Internet is an example of:

Internet इसका example है:

(A) LAN **(B)** MAN
(C) WAN **(D)** PAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q12]

✔ **Correct: (C) WAN** — The Internet is a global system of interconnected networks - it is the largest WAN (Wide Area Network).

Internet interconnected networks का एक global system है - यह सबसे बड़ा WAN (Wide Area Network) है।

★ **Nukta:** Internet = सबसे बड़ा WAN (network of networks).

Internet = सबसे बड़ा WAN (network of networks).

Q69. What is the difference between an 'internet' and the 'Internet'?

'internet' और 'Internet' में अंतर क्या है?

(A) There is no difference; both mean the World Wide Web / कोई अंतर नहीं; दोनों का अर्थ World Wide Web है **(B)** An internet is any network of networks; the Internet is the specific global one we use today / internet कोई भी network of networks है; Internet आज की वह विशिष्ट विश्वव्यापी network of networks है
(C) An internet is smaller than a LAN; the Internet is larger than a WAN / internet LAN से छोटा है; Internet WAN से बड़ा है **(D)** An internet uses cables; the Internet uses only wireless / internet cables use करता है; Internet सिर्फ wireless use करता है

[Source: Forouzan Ch1 (Q1-10) — Practice Set]

✔ **Correct: (B) An internet is any network of networks; the Internet is the specific global one we use today** — An 'internet' (lower-case) is any generic network of networks. The 'Internet' (capital I) is THE specific worldwide internet that grew from ARPANET.

'internet' (छोटा i) कोई भी generic network of networks है। 'Internet' (बड़ा I) वह विशिष्ट विश्वव्यापी internet है जो ARPANET से विकसित हुआ।

★ **Nukta:** internet = generic network of networks. Internet = THE global network (capital I).

internet = generic network of networks। Internet = वह global network (बड़ा I)।

Q70. SAN is mainly used for:
SAN का मुख्य उपयोग किसके लिए होता है?

(A) File Sharing **(B)** Storage Management
(C) Internet Access **(D)** Printing Documents / documents print करना

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q13]

✔ **Correct: (B) Storage Management** — SAN (Storage Area Network) is a dedicated high-speed private network used for storage management - connecting servers to storage devices in data centres.
SAN (Storage Area Network) एक dedicated high-speed private network है जो storage management के लिए use होता है - servers को storage devices से जोड़ने के लिए (data centres में).
★ **Nukta:** SAN = Storage Area Network - servers <-> storage devices, data centre.
SAN = Storage Area Network - servers <-> storage devices, data centre.

Q71. PAN is used for connecting:
PAN किसके connect करने के लिए use होता है?

(A) Personal devices like mobile, laptop, Bluetooth headset / personal devices जैसे mobile, laptop, Bluetooth headset **(B)** Computers in a company / एक company के computers
(C) Computers in a city / एक city के computers **(D)** Global offices

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q14]

✔ **Correct: (A) Personal devices like mobile, laptop, Bluetooth headset** — PAN (Personal Area Network) is a network organized around an individual person - connecting personal devices like mobile, laptop, Bluetooth headset.
PAN (Personal Area Network) एक व्यक्ति के around organized एक network है - personal devices जैसे mobile, laptop, Bluetooth headset को जोड़ने के लिए।
★ **Nukta:** PAN = Personal Area Network - एक person के devices (mobile, laptop, headset).
PAN = Personal Area Network - एक person के devices (mobile, laptop, headset).

Q72. A school has four computer labs in the same building. Each lab has 20 computers connected to a central server. Students can share files and print from any lab. Which type of network is implemented in this school?
Ek school में एक ही building में चार computer labs हैं. हर lab में 20 computers एक central server से जुड़े हैं। Students किसी भी lab से files share और print कर सकते हैं। School में कौन सा network implemented है?

(A) WAN **(B)** MAN
(C) LAN **(D)** CAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q15]

✔ **Correct: (C) LAN** — All four labs are in the same building and share a central server - this is a small, privately-owned, single-site network = LAN (Local Area Network).
Chaaron labs एक ही building में हैं और एक central server share करते हैं - यह एक छोटा, privately-owned, single-site network है = LAN (Local Area Network).
★ **Nukta:** LAN = single building/campus, private, high speed. Example: school lab, office.
LAN = एक building/campus, private, high speed. Example: school lab, office.

Q73. A university has different departments in separate buildings like Engineering, Arts, and Commerce. Each department has its own LAN. All these LANs are connected together for communication. Which type of network connects all these LANs?
Ek university में अलग-अलग departments अलग buildings में हैं जैसे Engineering, Arts, Commerce। हर department का apna LAN है। यह सब LANs communication के लिए आपस में जुड़े हैं। कौन सा network in सब LANs को जोड़ता है?

(A) LAN **(B)** CAN
(C) MAN **(D)** WAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q16]

✔ **Correct: (B) CAN** — A CAN (Campus Area Network) interconnects the LANs of different buildings across a campus (university/corporate). Here, multiple departmental LANs are joined = CAN.
CAN (Campus Area Network) एक campus (university/ corporate) की अलग buildings में maujood LANs को interconnect करता है। यहाँ कई departmental LANs जुड़े हैं = CAN.
★ **Nukta:** CAN = Campus Area Network - एक campus की multiple buildings के LANs को जोड़ता है.
CAN = Campus Area Network - एक campus की कई buildings के LANs को जोड़ता है।

Q74. An international bank connects its main office in New York with branches in Delhi, London, and Tokyo. Which network type is used here?

Ek international bank अपने main office (New York) को Delhi, London, और Tokyo की branches से जोड़ता है। यहाँ कौन सा network type use होता है?

(A) MAN **(B)** WAN
(C) CAN **(D)** LAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q17]

✔ **Correct: (B) WAN** — Branches in different cities and countries (New York, Delhi, London, Tokyo) are connected over long distances - this is a WAN (Wide Area Network).

Alag-अलग cities और deshon में branches (New York, Delhi, London, Tokyo) लंबी दूरी पर जुड़े हैं - यह WAN (Wide Area Network) है।

★ **Nukta:** WAN = long distance (city/country/continent). Bank branches worldwide = WAN.

WAN = लंबी दूरी (city/देश/continent). Worldwide bank branches = WAN.

Q75. A company has several servers that store and manage large volumes of data. All storage devices are interconnected for high-speed data access. Which type of network is this?

Ek company के पास कई servers हैं जो बड़ी matra में data store और manage करते हैं। सब storage devices high-speed data access के लिए interconnected हैं। यह कौन सा network है?

(A) LAN **(B)** SAN
(C) CAN **(D)** WAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q18]

✔ **Correct: (B) SAN** — Servers plus storage devices, all interconnected for high-speed data access = SAN (Storage Area Network), a dedicated network for storage.

Servers + storage devices, सब high-speed data access के लिए interconnected = SAN (Storage Area Network), storage के लिए एक dedicated network.

★ **Nukta:** SAN = Storage Area Network - storage devices को high-speed access dene के लिए.

SAN = Storage Area Network - storage devices को high-speed access dene के लिए।

Q76. A supplier accessing a company's order tracking system uses:

Ek supplier जो company के order tracking system को access करता है, कौन सा network use करता है?

(A) Internet **(B)** Intranet
(C) Extranet **(D)** VPN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q19]

✔ **Correct: (C) Extranet** — An Extranet is an intranet extended to authorized outsiders (suppliers, partners, customers). A supplier accessing the company's order tracking system = Extranet.

Extranet एक ऐसा intranet है जो authorized outsiders (suppliers, partners, customers) के लिए extend किया गया हो। Company के order tracking system को access करने वाला supplier = Extranet.

★ **Nukta:** Extranet = intranet + authorized outsiders (suppliers/partners/customers).

Extranet = intranet + authorized outsiders (suppliers/partners/customers).

Q77. Which network type provides maximum security and privacy?

कौन सा network type अधिकतम (maximum) security और privacy provide करता है?

(A) Internet **(B)** Intranet
(C) Extranet **(D)** None

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q20]

✔ **Correct: (B) Intranet** — An Intranet is a private network inside an organization, accessible only to its members - so it provides maximum security and privacy. Internet is public; extranet is shared with outsiders.

Intranet एक organization के अंदर का private network है, सिर्फ uske members के लिए accessible - इसलिए यह maximum security और privacy provide करता है। Internet public है; extranet outsiders के साथ share होता है।

★ **Nukta:** Security order: Intranet (सबसे secure/private) > Extranet > Internet (public).

Security order: Intranet (सबसे secure/private) > Extranet > Internet (public).

Q78. A software company shares project reports and progress updates with its clients through a secure online portal that requires login credentials. Which network type is this?

Ek software company अपने clients के साथ project reports और progress updates एक secure online portal के through share करती है जिसमें login credentials चाहिए। यह कौन सा network type है?

(A) Internet **(B)** Intranet
(C) Extranet **(D)** MAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q21]

✔ **Correct: (C) Extranet** — Sharing internal project information with clients (outsiders) through a secure login portal = Extranet (the company's intranet extended to authorized clients).

Clients (outsiders) के साथ internal project information share करना, एक secure login portal के through = Extranet (company का intranet authorized clients के लिए extend किया गया).

★ **Nukta:** Clients को secure portal से share करना = Extranet.

Clients को secure portal से share करना = Extranet.

Q79. Bank employees access internal financial databases, while customers access their accounts through online banking websites. Which networks are being used by employees and customers respectively?

Bank employees internal financial databases access करते हैं, jabki customers online banking websites के through अपने accounts access करते हैं। Employees और customers kramashah कौन से networks use कर रहे हैं?

(A) Employees - Intranet; Customers - Extranet **(B)** Employees - Internet; Customers - Intranet
(C) Employees - Extranet; Customers - Internet **(D)** Both - Internet / दोनों - Internet

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q22]

✔ **Correct: (A) Employees - Intranet; Customers - Extranet** — Employees use the bank's internal private network = Intranet. Customers access the bank's system from outside = Extranet (per the official key: the bank's online banking is its intranet extended to customers).

Employees bank के internal private network = Intranet use करते हैं। Customers बाहर से bank के system को access करते हैं = Extranet (official key के anusaar: bank का online banking uske intranet का customers के लिए extension है).

★ **Nukta:** Employees -> Intranet (internal). Outsiders (customers/suppliers) -> Extranet. Public -> Internet.

Employees -> Intranet (internal). Outsiders (customers/suppliers) -> Extranet. Public -> Internet.

Q80. Three or more devices share a link in _____ connection.

Teen या अधिक devices एक link _____ connection में share करते हैं।

(A) Unipoint **(B)** Multipoint
(C) Point to point **(D)** None of the mentioned / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q36]

✔ **Correct: (B) Multipoint** — A multipoint (multidrop) connection is one in which more than two devices share a single link; the capacity of the channel is shared spatially or temporally. (Forouzan)

Multipoint (multidrop) connection में do से ज़्यादा devices एक single link share करते हैं; channel की capacity spatially या temporally share होती है। (Forouzan)

★ **Nukta:** Multipoint = एक link, कई devices; capacity shared. Point-to-point = dedicated link, सिर्फ 2 devices.

Multipoint = एक link, कई devices; capacity shared. Point-to-point = dedicated link, सिर्फ 2 devices.

Q81. What are the two types of line configuration?

Line configuration के दो प्रकार क्या हैं?

(A) Simplex and Half-Duplex / Simplex और Half-Duplex **(B)** Serial and Parallel / Serial और Parallel
(C) Point-to-Point and Multipoint / Point-to-Point और Multipoint **(D)** Analog and Digital / Analog और Digital

[Source: Forouzan Ch1 (Q1-4) — Practice Set]

✔ **Correct: (C) Point-to-Point and Multipoint** — Forouzan: line configuration describes how devices are connected — a dedicated link between two devices (point-to-point) or a link shared by three or more devices (multipoint).

Forouzan: line configuration बताती है कि devices कैसे जुड़ी हैं — दो devices के बीच dedicated link (point-to-point) या तीन+ devices का shared link (multipoint)।

★ **Nukta:** Line configuration के सिर्फ 2 type: Point-to-Point & Multipoint. Simplex/Half/Full-Duplex TRANSMISSION MODES हैं.

Line configuration के 2 type: Point-to-Point और Multipoint। Simplex/Half/Full-Duplex transmission modes हैं।

Q82. What is the advantage of a multipoint connection over a point-to-point connection?

Point-to-point connection की तुलना में multipoint connection का क्या लाभ है?

(A) Fewer cables are needed, so the cost is lower / कम cables लगती हैं, इसलिए cost कम होती है **(B)** It provides a dedicated link for every pair of devices / हर device-युग्म को dedicated link मिलता है

(C) It is more secure than point-to-point / यह point-to-point से ज़्यादा secure है **(D)** It needs more cables and ports / इसमें ज़्यादा cables और ports लगते हैं

[Source: Forouzan Ch1 (Q1-3) — Practice Set]

✔ **Correct: (A) Fewer cables are needed, so the cost is lower** — In a multipoint connection three or more devices SHARE one link, so fewer cables are required and the cost is shared — its main advantage over dedicated point-to-point links.

Multipoint connection में तीन+ devices एक link share करती हैं, इसलिए cables कम लगती हैं और cost कम आती है — यही इसका मुख्य लाभ है।

★ **Nukta:** Multipoint = कम cable = कम kharch. Point-to-point = dedicated link (ज़्यादा cable, ज़्यादा privacy).

Multipoint = कम cable = कम खर्च। Point-to-point = dedicated link (ज़्यादा cable, ज़्यादा privacy)।

Q83. A _____ connection provides a dedicated link between two devices.

Ek _____ connection do devices के बीच एक dedicated link provide करता है।

(A) point-to-point **(B)** multipoint

(C) primary **(D)** secondary

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q42]

✔ **Correct: (A) point-to-point** — A point-to-point connection provides a dedicated link between two devices; the entire capacity of the link is reserved for transmission between those two devices. (Forouzan)

Point-to-point connection do devices के बीच एक dedicated link provide करता है; link की पूरी capacity सिर्फ unhi do devices के लिए reserved होती है। (Forouzan)

★ **Nukta:** Point-to-point = dedicated link, पूरी capacity सिर्फ 2 devices के लिए. Example: TV remote (IR), leased line.

Point-to-point = dedicated link, पूरी capacity सिर्फ 2 devices के लिए. Example: TV remote (IR), leased line.

Q84. How many point-to-point WANs are needed to connect n LANs so that every LAN can directly communicate with every other LAN?

n LANs को इस तरह जोड़ने के लिए कितने point-to-point WANs चाहिए कि हर LAN हर दूसरी LAN से सीधे communicate कर सके?

(A) n **(B)** n - 1

(C) n(n - 1)/2 **(D)** n^2

[Source: Forouzan Ch1 (Q1-13) — Practice Set]

✔ **Correct: (C) n(n - 1)/2** — Direct communication between every pair of LANs needs one WAN link per pair — n(n-1)/2 links (the same mesh formula).

हर LAN-युग्म के बीच सीधे communication के लिए हर युग्म को एक WAN link चाहिए — n(n-1)/2 links (mesh जैसा ही formula)।

★ **Nukta:** Full mesh of LANs = n(n-1)/2 WAN links. Formula mesh वाला ही है.

LANs का full mesh = n(n-1)/2 WAN links। Formula mesh जैसा ही है।

Q85. When a person makes a local telephone call to another person, what type of connection is it?

जब कोई व्यक्ति दूसरे व्यक्ति को local telephone call करता है, तो यह किस प्रकार का connection है?

(A) Multipoint connection **(B)** Point-to-point connection

(C) Broadcast connection **(D)** Multicast connection

[Source: Forouzan Ch1 (P1-7) — Practice Set (Problem)]

✔ **Correct: (B) Point-to-point connection** — A local telephone call creates a dedicated circuit between exactly two parties through the exchange — a point-to-point connection.

Local telephone call में exchange के medium से ठीक दो parties के बीच dedicated circuit बनता है — यह point-to-point connection है।

★ **Nukta:** Telephone call = 2 parties, dedicated link = point-to-point. Bus/Ethernet = multipoint.

Telephone call = 2 parties, dedicated link = point-to-point। Bus/Ethernet = multipoint।

Q86. Which of the following is an example of Client-Server architecture?

निम्नलिखित में से कौन Client-Server architecture का example है?

(A) Chat between two Bluetooth devices / do Bluetooth devices के बीच chat **(B)** Web browser accessing a website / web browser जो एक website access कर रहा है **(C)** File transfer via USB cable / USB cable से file transfer **(D)** Two laptops connected directly / do laptops directly connected

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q23]

✔ **Correct: (B) Web browser accessing a website** — A web browser (client) requesting and receiving a web page from a web server is the classic client-server architecture: the client requests a service and the server provides it.

Web browser (client) जो एक web server से web page request करता है और receive करता है - यह classic client-server architecture है: client service request करता है और server use provide करता है।

★ **Nukta:** Client-Server = client request करता है, server service provide करता है. Example: web browser - website.

Client-Server = client request करता है, server service provide करता है। Example: web browser - website.

Q87. Which of the following uses Peer-to-Peer (P2P) architecture, not Client-Server?

निम्नलिखित में से कौन Peer-to-Peer (P2P) architecture use करता है, Client-Server नहीं?

(A) BitTorrent file sharing **(B)** Google search **(C)** Gmail **(D)** YouTube

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q24]

✔ **Correct: (A) BitTorrent file sharing** — BitTorrent is peer-to-peer: users (peers) share file pieces directly with each other without a central server. Google search, Gmail and YouTube are client-server services.

BitTorrent peer-to-peer है: users (peers) file के tukde directly आपस में share करते हैं बिना किसी central server के। Google search, Gmail और YouTube client-server services हैं।

★ **Nukta:** P2P = peers आपस में directly share करते हैं (BitTorrent). Client-Server = server service देता है (Google, Gmail, YouTube).

P2P = peers आपस में directly share करते हैं (BitTorrent). Client-Server = server service देता है (Google, Gmail, YouTube).

Q88. In a bank, customers use ATMs to check balance or withdraw cash. ATM machines send the request to a central computer for verification and processing. Which type of architecture is used here?

Ek bank में customers ATMs से balance check या cash withdraw करते हैं। ATM machines request एक central computer को verification और processing के लिए bhejte हैं। यहाँ कौन सी architecture use होती है?

(A) Peer-to-Peer **(B)** Client-Server **(C)** Multipoint **(D)** Point-to-Point

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q25]

✔ **Correct: (B) Client-Server** — The ATM (client) sends a request to a central computer (server) for verification and processing - the client requests a service and the server provides it = client-server architecture.

ATM (client) request एक central computer (server) को verification और processing के लिए भेजता है - client service request करता है और server use provide करता है = client-server architecture.

★ **Nukta:** ATM -> central server = client-server (client request, server verify/process).

ATM -> central server = client-server (client request, server verify/process).

Q89. In a LAN, connectivity can be created using Wi-Fi or _____.

Ek LAN में connectivity Wi-Fi या _____ से बनाई जा सकती है।

(A) TV **(B)** Radio **(C)** Mobile **(D)** Ethernet

[Source: Repo: Random Test- Rajasthan PYQ (RPSC/RSSB)]

✔ **Correct: (D) Ethernet** — LAN connectivity is created using Wi-Fi (wireless) or Ethernet cables (wired). TV, radio and mobile are not LAN connectivity media.

LAN connectivity Wi-Fi (wireless) या Ethernet cables (wired) से बनाई जाती है। TV, radio और mobile LAN connectivity media नहीं हैं।

★ **Nukta:** LAN connectivity = Wi-Fi (wireless) + Ethernet (wired cable).

LAN connectivity = Wi-Fi (wireless) + Ethernet (wired cable).

Q90. Which network type is built on top of another existing network?

कौन-सा network type किसी मौजूदा network के ऊपर बनाया जाता है?

(A) Overlay network **(B)** Prime network
(C) Prior network **(D)** Chief network

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) Overlay network** — An overlay network is a virtual network built on top of another (host) network — e.g. VPN tunnels over the Internet or P2P networks over the Internet.

Overlay network एक virtual network है जो किसी दूसरे (host) network के ऊपर बनता है — जैसे Internet पर VPN tunnels या P2P networks।

★ **Nukta:** Overlay = network के ऊपर बना network (VPN classic example है). Baaki 3 asli network types नहीं हैं.

Overlay = network के ऊपर बना network (VPN इसका classic example है)। बाकी 3 real network types नहीं हैं।

Section F - Network History (ARPANET to Today's Network)

Syllabus: Network History - ARPANET to today's network

Q91. ARPANET stands for _____.

ARPANET का पूरा नाम है _____.

(A) Advanced Recheck Projects Agency Internet **(B)** Advanced Recheck Projects Agency Network
(C) Advanced Research Projects Agency Network
(D) Advanced Research Projects Agency Internet

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q26]

✔ **Correct: (C) Advanced Research Projects Agency Network** — ARPANET = Advanced Research Projects Agency Network. It was created by ARPA (Advanced Research Projects Agency) of the U.S. Department of Defense.

ARPANET = Advanced Research Projects Agency Network. इसे ARPA (Advanced Research Projects Agency) of U.S. Department of Defense ने बनाया था।

★ **Nukta:** ARPANET = Advanced Research Projects Agency Network (U.S. Dept. of Defense).

ARPANET = Advanced Research Projects Agency Network (U.S. Dept. of Defense).

Q92. The first network was:

सबसे पहला network था:

(A) CNNET **(B)** NSFNET
(C) ASAPNET **(D)** ARPANET

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q34]

✔ **Correct: (D) ARPANET** — The first network was ARPANET (1969), created by ARPA of the U.S. Department of Defense. NSFNET (1986) came later as a backbone.

सबसे पहला network ARPANET (1969) था, जिसे ARPA of U.S. Department of Defense ने बनाया था। NSFNET (1986) बाद में एक backbone के रूप में aaya था।

★ **Nukta:** First network = ARPANET (1969). NSFNET = 1986 (backbone).

First network = ARPANET (1969). NSFNET = 1986 (backbone).

Q93. In the original ARPANET, _____ were directly connected together.

Original ARPANET में, _____ आपस में directly connected the।

(A) IMPs / IMPs (Interface Message Processors) **(B)** host computers
(C) networks **(D)** routers

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q43]

✔ **Correct: (A) IMPs** — In the original ARPANET, the IMPs (Interface Message Processors - the packet switches) were directly connected together; the host computers connected to the IMPs.

Original ARPANET में, IMPs (Interface Message Processors - packet switches) आपस में directly connected the; host computers IMPs से connect होते the।

★ **Nukta:** Original ARPANET: IMPs (packet switches) directly connected; hosts attached to IMPs.

Original ARPANET: IMPs (packet switches) directly connected the; hosts IMPs से attach होते the।

Q94. The first operational computer network in the world was the _____ for the United States Department of Defense.

दुनिया का पहला operational computer network, United States Department of Defense के लिए _____ था।

(A) ARPANET **(B)** ERNET
(C) SKYNET **(D)** DARPANET

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q48]

✔ **Correct: (A) ARPANET** — ARPANET was the first operational computer network in the world, created for the U.S. Department of Defense (1969).

ARPANET दुनिया का पहला operational computer network था, जो U.S. Department of Defense के लिए बनाया गया था (1969).

★ **Nukta:** First operational network = ARPANET (1969, U.S. DoD).

First operational network = ARPANET (1969, U.S. DoD).

Q95. ARPANET - the precursor of today's Internet - was created in _____ by ARPA (Advanced Research Projects Agency) of the U.S. Department of Defense, starting with four nodes (UCLA, SRI, UC Santa Barbara, University of Utah).

ARPANET - aaj के Internet का purvaj - _____ में ARPA (Advanced Research Projects Agency), U.S. Department of Defense द्वारा बनाया गया था, चार nodes से shuruaat हुई (UCLA, SRI, UC Santa Barbara, University of Utah).

(A) 1957 **(B)** 1969
(C) 1983 **(D)** 1991

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (B) 1969** — ARPANET was born in 1969 with four nodes: UCLA, Stanford Research Institute (SRI), UC Santa Barbara and the University of Utah. The first message was sent on 29 October 1969 (the LO of LOGIN).

ARPANET 1969 में चार nodes के साथ बना था: UCLA, Stanford Research Institute (SRI), UC Santa Barbara और University of Utah. पहला message 29 October 1969 को भेजा गया था (LOGIN का LO).

★ **Nukta:** ARPANET = 1969, 4 nodes (UCLA, SRI, UCSB, Utah); first message 29 Oct 1969 (LO of LOGIN).

ARPANET = 1969, 4 nodes (UCLA, SRI, UCSB, Utah); पहला message 29 Oct 1969 (LOGIN का LO).

Q96. TCP/IP (developed by Vint Cerf and Bob Kahn) was adopted by ARPANET as its official protocol suite on 1 January _____ - an event known as the fflag day.

TCP/IP (Vint Cerf और Bob Kahn द्वारा develop किया गया) को ARPANET ने apna official protocol suite बनाया था 1 January _____ को - is event को fflag day कहा जाता है।

(A) 1969 **(B)** 1972
(C) 1983 **(D)** 1995

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (C) 1983** — On 1 January 1983, ARPANET officially switched to TCP/IP - the fflag day - making TCP/IP the common language of the network. Cerf and Kahn developed TCP/IP in the 1970s (publicly demonstrated in 1972).

1 January 1983 को ARPANET officially TCP/IP पर switch हो गया - fflag day - जिससे TCP/IP network की common language बन गया। Cerf और Kahn ने TCP/IP 1970s में develop किया था (1972 में publicly demonstrate हुआ).

★ **Nukta:** Fflag day = 1 Jan 1983 (ARPANET -> TCP/IP). TCP/IP developed by Cerf & Kahn in 1970s (demo 1972). Syllabus का TCP/IP model 1972 = development era.

Fflag day = 1 Jan 1983 (ARPANET -> TCP/IP). TCP/IP Cerf & Kahn ने 1970s में develop किया (demo 1972). Syllabus का TCP/IP model 1972 = development era.

Q97. NSFNET, created by the U.S. National Science Foundation in _____, became a major backbone of the Internet, connecting universities and research centres.

NSFNET, U.S. National Science Foundation द्वारा _____ में बनाया गया, Internet का एक major backbone बन गया, universities और research centres को जोड़ता था।

(A) 1969 **(B)** 1983
(C) 1986 **(D)** 1991

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (C) 1986** — NSFNET was created in 1986 by the National Science Foundation; it connected universities and research centres with T1 lines (1.544 Mbps) and became a major Internet backbone.

NSFNET 1986 में National Science Foundation द्वारा बनाया गया था; isne universities और research centres को T1 lines (1.544 Mbps) से जोड़ा और एक major Internet backbone बन गया।

★ **Nukta:** NSFNET = 1986 (National Science Foundation), T1 backbone -> Internet backbone.

NSFNET = 1986 (National Science Foundation), T1 backbone -> Internet backbone.

Q98. The World Wide Web (WWW) was invented in 1991 at CERN by _____, and it made the Internet popular among the general public.

World Wide Web (WWW) का invention 1991 में CERN में _____ ने किया था, और isne Internet को आम public के बीच popular बनाया।

(A) Vint Cerf **(B)** Tim Berners-Lee
(C) Bob Kahn **(D)** Marc Andreessen

[Source: Extra (Forouzan) - fills syllabus gap]

✔ **Correct: (B) Tim Berners-Lee** — Tim Berners-Lee invented the WWW at CERN in 1991 (using HTML, HTTP and URLs). The Mosaic browser (1993) later made the web popular; the Internet opened to commercial use in the 1990s.

Tim Berners-Lee ने 1991 में CERN में WWW का invention किया था (HTML, HTTP और URLs का use karke). Mosaic browser (1993) ने बाद में web को popular बनाया; Internet 1990s में commercial use के लिए khula।

★ **Nukta:** WWW = 1991, Tim Berners-Lee @ CERN. Mosaic browser 1993. Vint Cerf & Bob Kahn = fathers of the Internet (TCP/IP).

WWW = 1991, Tim Berners-Lee @ CERN. Mosaic browser 1993. Vint Cerf & Bob Kahn = Internet के fathers (TCP/IP).

Section G - General Network Basics (WWW, E-mail, Protocols, Servers, Switching - from repo intro tests)

Syllabus: Introduction of DC - general basics asked in BCI/RPSC exams (Specific Address/URL, Application services, Switching)

Q99. A collection of hyperlinked documents on the internet forms the?

Internet पर hyperlinked documents का sangrah (collection) क्या banata है?

(A) World Wide Web (WWW) **(B)** E-mail system
(C) Mailing list **(D)** Hypertext markup language / hypertext markup language (HTML)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q5]

✔ **Correct: (A) World Wide Web (WWW)** — A collection of hyperlinked documents on the Internet forms the World Wide Web (WWW). HTML is the markup language used to create those documents.

Internet पर hyperlinked documents का sangrah World Wide Web (WWW) banata है। HTML वह markup language है जिस से यह documents bante हैं।

★ **Nukta:** WWW = hyperlinked documents का collection. HTML = un documents की language (WWW नहीं).

WWW = hyperlinked documents का collection. HTML = un documents की language है (WWW नहीं).

Q100. Which one of the following is a valid email address?

निम्नलिखित में से कौन सा एक valid email address है?

(A) computech@academy.com **(B)** gmail.com
(C) compu@.com **(D)** computech@books

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q6]

✔ **Correct: (A) computech@academy.com** — A valid email address has a local part, the @ symbol, and a domain with a top-level domain (TLD). computech@academy.com follows this format.

Ek valid email address में local part, @ symbol, और एक domain with top-level domain (TLD) होता है। computech@academy.com यह format follow करता है।

★ **Nukta:** Valid email = local@domain.TLD (e.g., name@site.com).

Valid email = local@domain.TLD (जैसे name@site.com).

Q101. Which of the following best describes uploading information?

निम्नलिखित में से कौन uploading information को सबसे achhe से describe करता है?

(A) Sorting data on a disk drive / disk drive पर data को sort करना **(B)** Sending information to a host computer / information को एक host computer को भेजना
(C) Receiving information from a host computer / एक host computer से information receive करना **(D)** Sorting data on a hard drive / hard drive पर data को sort करना

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q7]

✔ **Correct: (B) Sending information to a host computer** — Uploading means sending/transferring information from a local device to a host/remote computer over a network.

Uploading का मतलब है information को एक local device से एक host/remote computer को network के through भेजना।

★ **Nukta:** Uploading = local से remote/host को भेजना. Downloading = remote से local को lena.

Uploading = local से remote/host को भेजना. Downloading = remote से local को lena.

Q102. Which one of the following is the most common internet protocol?

Nimnlikhis में से कौन सा सबसे common internet protocol है?

(A) HTML **(B)** NetBEUI
(C) TCP/IP **(D)** IPX/SPX

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q8]

✔ **Correct: (C) TCP/IP** — TCP/IP is the core protocol suite of the Internet - it is the most common protocol used on the Internet. HTML is a markup language, not a protocol.

TCP/IP Internet का core protocol suite है - यह Internet पर सबसे common protocol है। HTML एक markup language है, protocol नहीं।

★ **Nukta:** Internet का core protocol = TCP/IP. HTML = language, protocol नहीं.

Internet का core protocol = TCP/IP. HTML = language है, protocol नहीं.

Q103. The dial-up connections to access the internet use ___ protocol.

Internet access के लिए dial-up connections ___ protocol use करते हैं।

(A) PPP (Point to Point protocol) **(B)** X.25 protocol
(C) PLIP protocol **(D)** None of the above / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q18]

✔ **Correct: (A) PPP (Point to Point protocol)** — Dial-up connections to access the Internet use PPP (Point-to-Point Protocol), which carries IP traffic over the telephone line.

Internet access के लिए dial-up connections PPP (Point-to-Point Protocol) use करते हैं, जो telephone line पर IP traffic carry करता है।

★ **Nukta:** Dial-up Internet = PPP (Point-to-Point Protocol).

Dial-up Internet = PPP (Point-to-Point Protocol).

Q104. When a resident uses a dial-up or DSL service to connect to the Internet, what is the role of the telephone company?

जब कोई व्यक्ति Internet से जुड़ने के लिए dial-up या DSL service use करता है, तो telephone company की भूमिका क्या होती है?

(A) It provides the switching/connection path from the user to the ISP / वह user से ISP तक switching/connection path देती है **(B)** It hosts every website the user visits / वह user की हर website host करती है
(C) It assigns IP addresses to every website / वह हर website को IP address देती है **(D)** It encrypts all the user's data / वह user का सारा data encrypt करती है

[Source: Forouzan Ch1 (Q1-14) — Practice Set]

✔ **Correct: (A) It provides the switching/ connection path from the user to the ISP** — Forouzan: with dial-up/DSL the telephone company is a switching service provider — it provides the local loop and connection path (circuit) from the customer to the ISP; the ISP connects to the Internet.

Forouzan: dial-up/DSL में telephone company switching service provider है — वह customer से ISP तक local loop/ connection path देती है; ISP Internet से जोड़ता है।

★ **Nukta:** Telephone company सिर्फ 'road' देती है (local loop + switching). Website hosting/IP assignment ISP/DNS का काम है.

Telephone company सिर्फ रास्ता देती है (local loop + switching)। Website hosting/IP assignment ISP/DNS का काम है।

Q105. Home broadband connections that work on DSL (Digital Subscriber Line) use ___ protocol.

DSL (Digital Subscriber Line) पर काम करने वाले home broadband connections ___ protocol use करते हैं।

(A) PPPoE (PPP over Ethernet) **(B)** PPPoA (PPP over ATM)
(C) Bridge protocol **(D)** All of the above / ऊपर Sabhi

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q19]

✔ **Correct: (A) PPPoE (PPP over Ethernet)** — Home DSL broadband connections typically use PPPoE (PPP over Ethernet) to carry PPP traffic over the DSL/Ethernet connection.

Home DSL broadband connections आम तौर पर PPPoE (PPP over Ethernet) use करते हैं ताकि PPP traffic DSL/Ethernet connection पर carry हो sake।

★ **Nukta:** Home DSL broadband = PPPoE (PPP over Ethernet).

Home DSL broadband = PPPoE (PPP over Ethernet).

Q106. Which of the following servers allows LAN users to share data?

निम्नलिखित में से कौन सा server LAN users को data share करने की अनुमति देता है?

(A) Data server **(B)** Point server
(C) File server **(D)** Communication server

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q22]

✔ **Correct: (C) File server** — A file server stores files and allows LAN users to share/access data (files) over the network.

Ek file server files store करता है और LAN users को network पर data (files) share/access करने की अनुमति देता है।

★ **Nukta:** File server = files store + LAN users को share करने देता है.

File server = files store करता है + LAN users को share करने देता है।

Q107. EDI stands for ____.
EDI का पूरा नाम है ____.

(A) Electronic Data Interchange **(B)** Electronic Digital Internet
(C) Electronic Digital Interchange **(D)** Electronic Data Internet

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q25]

✔ **Correct: (A) Electronic Data Interchange** — EDI = Electronic Data Interchange - the computer-to-computer exchange of business documents (orders, invoices) in a stअंदरd electronic format.
EDI = Electronic Data Interchange - business documents (orders, invoices) का computer-to-computer exchange, एक stअंदरd electronic format में.
★ **Nukta:** EDI = Electronic Data Interchange (business documents का e-exchange).
EDI = Electronic Data Interchange (business documents का e-exchange).

Q108. A computer network permits sharing of:
Ek computer network kis cheez की sharing की अनुमति देता है?

(A) Resources / resources (printer, disk, etc.) **(B)** Information / information (data)
(C) Both a & b / दोनों a और b **(D)** None of these / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q47 / CN Topic Test-02 (UNIT TEST-2), Q47]

✔ **Correct: (C) Both a & b** — A computer network permits sharing of both resources (printers, storage, devices) and information (data/messages) among connected users.
Ek computer network दोनों cheezon की sharing की अनुमति देता है: resources (printers, storage, devices) और information (data/messages), connected users के बीच।
★ **Nukta:** Network = resources + information sharing.
Network = resources + information sharing.

Q109. How many digits of the Data Network Identification Code (DNIC) identify the country?
Data Network Identification Code (DNIC) के kitne digits देश (country) की पहचान करते हैं?

(A) first three / पहले तीन **(B)** first four / पहले चार
(C) first five / पहले पाँच **(D)** first six / पहले छह

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q15]

✔ **Correct: (A) first three** — In the DNIC (used in X.121 addressing), the first three digits are the country code and the fourth digit identifies the network within that country. So the first three digits identify the country.
DNIC (X.121 addressing में use होता है) में पहले तीन digits country code होते हैं और चौथा digit उस देश के अंदर के network की पहचान करता है। इसलिए पहले तीन digits देश की पहचान करते हैं।
★ **Nukta:** DNIC: first 3 digits = country code, 4th digit = network code.
DNIC: पहले 3 digits = country code, चौथा digit = network code.

Q110. Consider the switching statements: 1. Circuit Switching establishes a dedicated path. 2. Packet Switching routes data in packets on different paths. 3. Message Switching uses store-and-forward. Which statements are correct?
Switching statements देखिए: 1. Circuit Switching dedicated path बनाता है। 2. Packet Switching data को packets में अलग paths पर भेजता है। 3. Message Switching store-and-forward use करता है। कौन से statements सही हैं?

(A) 1 and 2 only / सिर्फ 1 और 2 **(B)** 2 and 3 only / सिर्फ 2 और 3
(C) 1 and 3 only / सिर्फ 1 और 3 **(D)** 1, 2 and 3 / 1, 2 और 3

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (D) 1, 2 and 3** — All three correct: circuit switching sets up a dedicated end-to-end path; packet switching splits data into packets that may take different paths; message switching stores the whole message at each node and forwards it (store-and-forward).
तीनों सही: circuit switching dedicated end-to-end path बनाता है; packet switching data को packets में बाँटकर अलग paths से भेजता है; message switching हर node पर पूरा message store करके आगे भेजता है।
★ **Nukta:** 3 switching types: Circuit (dedicated path), Packet (packets, अलग paths), Message (store-and-forward) — teeno statements correct.
3 switching types: Circuit (dedicated path), Packet (packets, अलग paths), Message (store-and-forward) — तीनों सही।

Q111. All the information on the internet is passed using _____ switching method.

Internet पर saari information _____ switching method से pass होती है।

(A) Portable **(B)** Packets
(C) Pouch **(D)** Primary

[Source: Repo: Random Test- Rajasthan PYQ (RPSC/RSSB)]

✔ **Correct: (B) Packets** — The Internet uses packet switching: data is broken into small packets that travel independently over the network and are reassembled at the destination.

Internet packet switching use करता है: data को chhote packets में tod कर भेजा जाता है जो network पर independently travel करते हैं और destination पर reassemble होते हैं।

★ **Nukta:** Internet = packet switching. Circuit switching = dedicated path (old telephone).

Internet = packet switching. Circuit switching = dedicated path (पुराना telephone).

Q112. Which technique breaks data into packets for transmission?

Transmission के लिए data को packets में तोड़ने की technique कौन-सी है?

(A) Circuit switching **(B)** Packet switching
(C) Frequency division multiplexing **(D)** Time division multiplexing

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)]

✔ **Correct: (B) Packet switching** — Packet switching breaks the message into packets, sends each (possibly over different paths) and reassembles them at the destination. Circuit switching sends a continuous stream over a dedicated path; FDM/TDM are multiplexing techniques.

Packet switching message को packets में तोड़ता है, हर packet भेजता है और destination पर जोड़ता है। Circuit switching dedicated path पर continuous stream भेजता है; FDM/TDM multiplexing techniques हैं।

★ **Nukta:** Internet packet switching पर चलता है — data packets में tootta है.

Internet packet switching पर चलता है — data packets में टूटता है।

Q113. Which of the following is true for Circuit Switching handling multiple simultaneous connections?

एक साथ अनेक connections संभालते समय Circuit Switching के बारे में क्या सही है?

(A) A single path is used and shared for all data / एक ही path सभी data के लिए share होता है **(B)** All connections are multiplexed into a single channel / सभी connections एक channel में multiplex होते हैं
(C) It encodes all data into a single channel / यह सारे data को एक channel में encode करता है **(D)** Separate physical paths are allocated for each channel / हर channel के लिए अलग physical paths दिए जाते हैं

[Source: RPSC Programmer 2024 Paper-1 (with official key)]

✔ **Correct: (D) Separate physical paths are allocated for each channel** — In circuit switching each connection gets its own dedicated physical path for the whole duration — multiple simultaneous connections mean multiple separate paths (as in the traditional telephone network). (Matches the official RPSC key.)

Circuit switching में हर connection को पूरे समय के लिए अपना dedicated physical path मिलता है — एक साथ अनेक connections = अलग-अलग paths (पारंपरिक telephone network जैसा)। (Official RPSC key से match.)

★ **Nukta:** Circuit switching = हर connection का apna अलग path. Packet switching = shared medium, packets अलग paths से.

Circuit switching = हर connection का अपना अलग path। Packet switching = shared medium, packets अलग paths से।

★ Nukte Rapid-Fire — Exam Pointers (quick revision)

One-line question (left) with its exam answer + trick (right). Revise these before the exam.

5 components of data communication? Data communication के 5 components?	✔ Message, Sender, Receiver, Medium, Protocol Message, Sender, Receiver, Medium, Protocol
3 criteria of a good network? अच्छे network के 3 criteria?	✔ Performance, Reliability, Security Performance, Reliability, Security
4 characteristics of data communication? Data communication के 4 characteristics?	✔ Delivery, Accuracy, Timeliness, Jitter Delivery, Accuracy, Timeliness, Jitter
Jitter means? Jitter का अर्थ?	✔ Delay variation between packets Packets के बीच delay का बदलाव
Protocol elements? Protocol के elements?	✔ Syntax, Semantics, Timing Syntax, Semantics, Timing
Simplex example? Simplex का उदाहरण?	✔ TV broadcast / keyboard→monitor (one-way) TV broadcast / keyboard→monitor (one-way)
Half-Duplex example? Half-Duplex का उदाहरण?	✔ Walkie-talkie (both ways, one at a time) Walkie-talkie (दोनों तरफ़, पर एक समय में एक)
Full-Duplex example? Full-Duplex का उदाहरण?	✔ Telephone (both ways, same time) Telephone (दोनों तरफ़, एक साथ)
Mesh cables formula? Mesh cables का formula?	✔ n(n-1)/2 cables, n-1 ports per device n(n-1)/2 cables, n-1 ports per device
Star / Ring / Bus cables? Star / Ring / Bus cables?	✔ Star = n, Ring = n, Bus = 1 Star = n, Ring = n, Bus = 1
Most fault-tolerant topology? सबसे ज़्यादा fault-tolerant topology?	✔ Mesh (redundant paths) Mesh (redundant paths)
Easy to install & reconfigure? Install/reconfigure में आसान?	✔ Star topology Star topology
Multipoint topology? Multipoint topology?	✔ Bus (shared backbone) Bus (shared backbone)
IEEE 802.2 / 802.3 / 802.5 / 802.11? IEEE 802.2 / 802.3 / 802.5 / 802.11?	✔ LLC / Ethernet(MAC) / Token Ring / Wi-Fi LLC / Ethernet(MAC) / Token Ring / Wi-Fi
Network size order? Network size का क्रम?	✔ PAN < LAN < MAN < WAN PAN < LAN < MAN < WAN
internet vs Internet? internet बनाम Internet?	✔ Any network of networks vs THE global one कोई भी network of networks बनाम वह विश्वव्यापी
Intranet vs Extranet? Intranet बनाम Extranet?	✔ Intranet = inside the organisation; Extranet = trusted outsiders (supplier) Intranet = संस्था के अंदर; Extranet = भरोसेमंद बाहरी (supplier)
Client-Server vs P2P examples? Client-Server बनाम P2P उदाहरण?	✔ ATM = Client-Server; BitTorrent = P2P ATM = Client-Server; BitTorrent = P2P
ARPANET facts? ARPANET तथ्य?	✔ 1969 (ARPA); TCP/IP adopted 1983; WWW 1991 (CERN, Tim Berners-Lee) 1969 (ARPA); TCP/IP adopted 1983; WWW 1991 (CERN, Tim Berners-Lee)
Internet switching? Internet कौन-सी switching use करता है?	✔ Packet switching (data travels in packets) Packet switching (data packets में चलता है)
Circuit switching key? Circuit switching की कुंजी?	✔ Dedicated separate physical path per connection हर connection का dedicated अलग physical path